8086 / 8088
Microprocessor Architecture

Comprehensive MCQ Question Bank

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Introduction to 8086

78 questions

Q1 EASY FREQUENTLY ASKED

The Intel 8086 microprocessor was introduced in which year?

A 1974

B 1976

C 1978

D 1980

Answer: C. 1978

Intel 8086 was introduced in 1978. It was the first 16-bit microprocessor from Intel.

Q2 EASY FREQUENTLY ASKED

The 8086 microprocessor is a ___-bit processor.

A 8-bit

B 16-bit

C 32-bit

D 4-bit

Answer: B. 16-bit

8086 is a 16-bit microprocessor — it processes 16 bits of data at a time.

Q3 EASY FREQUENTLY ASKED

How many address lines does the 8086 microprocessor have?

A 16

B 20

C 24

D 32

Answer: B. 20

8086 has 20 address lines (A0–A19), allowing it to address up to 2²■ = 1 MB of memory.

Q4 EASY FREQUENTLY ASKED

What is the maximum memory that the 8086 can address?

A 64 KB
B 256 KB
C 512 KB
D 1 MB

Answer: D. 1 MB
With 20 address lines, 8086 can address 2²■ = 1,048,576 bytes = 1 MB.

Q5 EASY FREQUENTLY ASKED

The data bus width of 8086 is:

A 8-bit
B 16-bit
C 32-bit
D 64-bit

Answer: B. 16-bit
8086 has a 16-bit data bus, allowing it to transfer 16 bits simultaneously.

Q6 MODERATE

The 8086 microprocessor has how many pins?

A 28
B 36
C 40
D 48

Answer: C. 40
The Intel 8086 is a 40-pin DIP (Dual Inline Package) IC.

Q7 MODERATE

What is the clock frequency of the original Intel 8086?

A. 1 MHz
B. 5 MHz
C. 8 MHz
D. 10 MHz

Answer: B. 5 MHz
The original 8086 operated at 5 MHz. Later versions (8086-2) ran at 8 MHz, and 8086-1 at 10 MHz.

Q8 EASY

Which company manufactured the Intel 8086?

A. AMD
B. Motorola
C. Intel
D. Zilog

Answer: C. Intel
Intel Corporation designed and manufactured the 8086 microprocessor in 1978.

Q9 MODERATE

The 8086 microprocessor is based on which architecture?

A. Harvard Architecture
B. Von Neumann Architecture
C. RISC Architecture
D. VLIW Architecture

Answer: B. Von Neumann Architecture
8086 uses Von Neumann architecture where program and data share the same memory and bus.

Q10 MODERATE FREQUENTLY ASKED

How many general-purpose registers does the 8086 have?

A. 4
B. 8
C. 16
D. 6

Answer: A. 4
8086 has 4 general-purpose registers: AX, BX, CX, DX (each 16-bit, splittable into H and L bytes).

Q11 EASY

Which register in 8086 is used as the accumulator?

A. BX
B. CX
C. AX
D. DX

Answer: C. AX
AX (Accumulator Register) is the primary register used for arithmetic and I/O operations.

Q12 EASY

The BX register in 8086 is also called:

A. Accumulator
B. Base Register
C. Counter Register
D. Data Register

Answer: B. Base Register
BX is called the Base Register and is commonly used as a memory base address pointer.

Q13 EASY

CX register in 8086 is primarily used as:

A. Accumulator
B. Base Register
C. Counter Register
D. Data Register

Answer: C. Counter Register
CX is the Counter Register, used in LOOP instructions and repetitive string operations.

Q14 MODERATE FREQUENTLY ASKED

DX register in 8086 is used for:

A. Loop counting only
B. I/O port addressing and 32-bit multiplication/division
C. Memory base addressing
D. Instruction pointer

Answer: B. I/O port addressing and 32-bit multiplication/division
DX (Data Register) holds I/O port address and upper 16 bits in 32-bit multiply/divide.

Q15 MODERATE FREQUENTLY ASKED

How many segment registers are present in 8086?

A. 2
B. 4
C. 6
D. 8

Answer: B. 4
8086 has 4 segment registers: CS (Code), DS (Data), SS (Stack), ES (Extra).

Q16 EASY

The CS register in 8086 stands for:

A Counter Segment

B Code Segment

C Control Segment

D Cache Segment

Answer: B. Code Segment
CS = Code Segment register. It holds the base address of the memory segment containing program code.

Q17 MODERATE FREQUENTLY ASKED

The IP (Instruction Pointer) in 8086 always points to:

A The current executing instruction

B The next instruction to be fetched

C The last executed instruction

D The stack top

Answer: B. The next instruction to be fetched
IP holds the offset address of the NEXT instruction to be fetched from memory by the BIU.

Q18 MODERATE FREQUENTLY ASKED

How many flag bits does the 8086 flag register have?

A 8

B 9

C 16

D 13

Answer: B. 9
8086 has 9 active flag bits: 6 status flags (CF, PF, AF, ZF, SF, OF) + 3 control flags (TF, IF, DF).

Q19 MODERATE

Which flag in 8086 is used to enable or disable maskable interrupts?

A Trap Flag

B Direction Flag

C Interrupt Flag

D Overflow Flag

Answer: C. Interrupt Flag
The Interrupt Flag (IF): IF=1 enables, IF=0 disables maskable hardware interrupts (INTR pin).

Q20 MODERATE

The 8086 operates in how many modes?

A One

B Two

C Three

D Four

Answer: B. Two
8086 operates in two modes: Minimum Mode (single processor) and Maximum Mode (multiprocessor).

Q21 DIFFICULT

In 8086, Minimum Mode is selected when MN/MX pin is:

A Low (0)

B High (1)

C Floating

D Any value

Answer: B. High (1)
MN/MX = 1 (tied to Vcc) selects Minimum Mode; MN/MX = 0 (tied to GND) selects Maximum Mode.

Q22 MODERATE FREQUENTLY ASKED

The instruction queue in 8086 can hold how many bytes?

A. 4 bytes
B. 6 bytes
C. 8 bytes
D. 16 bytes

Answer: B. 6 bytes
8086 has a 6-byte instruction prefetch queue, fetched ahead by BIU while EU executes.

Q23 MODERATE FREQUENTLY ASKED

The physical address in 8086 is formed by:

A. Segment Register + Offset
B. Segment Register × 10H + Offset
C. Segment Register + IP
D. Segment Register × 16 – Offset

Answer: B. Segment Register × 10H + Offset
Physical Address = Segment Register × 10H (left-shift 4 bits) + Offset (Effective Address).

Q24 MODERATE FREQUENTLY ASKED

The size of each segment in 8086 memory is a maximum of:

A. 512 bytes
B. 1 KB
C. 64 KB
D. 1 MB

Answer: C. 64 KB
Each memory segment in 8086 can be a maximum of 64 KB (65,536 bytes) in size.

Q25 EASY

The 8086 microprocessor belongs to which family?

A x64 family

B x86 family

C MIPS family

D ARM family

Answer: B. x86 family
8086 is the founding member of Intel's x86 family of processors.

Q26 EASY FREQUENTLY ASKED

What does BIU stand for in 8086 architecture?

A Binary Interface Unit

B Bus Interface Unit

C Basic Instruction Unit

D Buffer Interface Unit

Answer: B. Bus Interface Unit
BIU = Bus Interface Unit. It manages all bus operations: fetching instructions and transferring data.

Q27 EASY FREQUENTLY ASKED

What does EU stand for in 8086 architecture?

A External Unit

B Execution Unit

C Extended Unit

D Effective Unit

Answer: B. Execution Unit
EU = Execution Unit. It decodes and executes instructions handed over from the instruction queue.

Q28 MODERATE

The carry flag (CF) in 8086 is set when:

A. Result is zero

B. Result is negative

C. Carry out of MSB occurs in arithmetic

D. Signed overflow occurs

Answer: C. Carry out of MSB occurs in arithmetic
CF is set when there is a carry out of (or borrow into) the most significant bit during arithmetic.

Q29 EASY

The Zero Flag (ZF) in 8086 is set when:

A. Result is positive

B. Result of operation is zero

C. Result overflows

D. Result is negative

Answer: B. Result of operation is zero
ZF = 1 whenever the result of an ALU operation is zero.

Q30 MODERATE FREQUENTLY ASKED

Which of the following is NOT a status flag in 8086?

A. CF

B. ZF

C. DF

D. SF

Answer: C. DF
DF (Direction Flag) is a CONTROL flag, not a status flag. CF, ZF, SF are status flags.

Q31 EASY

The SP register in 8086 stands for:

A Stack Pointer

B Segment Pointer

C Source Pointer

D Special Purpose

Answer: A. Stack Pointer
SP = Stack Pointer. It holds the offset address of the current top of the stack.

Q32 MODERATE

How many index registers does 8086 have?

A 1

B 2

C 3

D 4

Answer: B. 2
8086 has 2 index registers: SI (Source Index) and DI (Destination Index).

Q33 MODERATE

The 8086 microprocessor uses which fabrication technology?

A TTL

B CMOS

C NMOS

D BiCMOS

Answer: C. NMOS
Intel 8086 uses NMOS (N-channel Metal-Oxide Semiconductor) technology.

Q34 DIFFICULT

The number of transistors in 8086 is approximately:

A. 6,500
B. 29,000
C. 134,000
D. 275,000

Answer: B. 29,000
Intel 8086 contains approximately 29,000 transistors.

Q35 MODERATE

The 8086 supports pipelining with how many stages?

A. 1
B. 2
C. 4
D. 5

Answer: B. 2
8086 has a 2-stage pipeline: BIU fetches instructions into queue while EU executes current instruction.

Q36 MODERATE

The 8086 supports how many I/O port addresses?

A. 256
B. 1024
C. 65536
D. 1048576

Answer: C. 65536
8086 can address 65,536 (64K) I/O ports using its 16-bit I/O address bus.

Q37 EASY

The 8086 instruction set is classified as:

A RISC

B CISC

C VLIW

D EPIC

Answer: B. CISC
8086 uses CISC (Complex Instruction Set Computing) — many complex, variable-length instructions.

Q38 MODERATE FREQUENTLY ASKED

The ALE signal in 8086 stands for:

A Address Latch Enable

B Arithmetic Logic Enable

C Address Line Enable

D Asynchronous Latch Enable

Answer: A. Address Latch Enable
ALE (Address Latch Enable) is a high pulse used to demultiplex address from AD0–AD15 lines.

Q39 MODERATE

In 8086, the BP register is used to:

A Point to top of stack

B Access stack frame parameters and local variables

C Hold I/O address

D Count loop iterations

Answer: B. Access stack frame parameters and local variables
BP (Base Pointer) is used to access function parameters and local variables in the stack frame.

Q40 MODERATE

Which 8086 register is used with DI for string operations as destination?

A. CS
B. DS
C. SS
D. ES

Answer: D. ES
ES (Extra Segment) is used with DI as the default destination segment in string operations.

Q41 MODERATE FREQUENTLY ASKED

The total number of registers in 8086 is:

A. 8
B. 12
C. 14
D. 16

Answer: C. 14
8086 has 14 programmer-accessible registers: 4 GP, 4 segment, SP, BP, SI, DI, IP, Flags.

Q42 DIFFICULT

The process name for 8086 fabrication was:

A. 3-micron NMOS
B. 3-micron HMOS
C. 2-micron CMOS
D. 5-micron NMOS

Answer: B. 3-micron HMOS
Intel 8086 was originally fabricated in HMOS (High-density N-channel MOS) 3-micron process technology.

Q43 MODERATE FREQUENTLY ASKED

The 8086 can handle how many interrupt types?

A. 64
B. 128
C. 256
D. 512

Answer: C. 256
8086 supports 256 interrupt types (INT 0 to INT 255), each with a 4-byte vector in the IVT.

Q44 MODERATE FREQUENTLY ASKED

Where is the Interrupt Vector Table (IVT) stored in 8086?

A. At top of memory (FFFF0H)
B. At the first 1024 bytes (00000H–003FFH)
C. In segment 8000H
D. In the stack segment

Answer: B. At the first 1024 bytes (00000H–003FFH)
The IVT occupies the lowest 1 KB of memory (00000H to 003FFH): 256 vectors × 4 bytes each.

Q45 MODERATE FREQUENTLY ASKED

Each interrupt vector in 8086 IVT contains:

A. 2 bytes (IP only)
B. 4 bytes (IP:CS)
C. 8 bytes
D. 1 byte

Answer: B. 4 bytes (IP:CS)
Each IVT entry = 4 bytes: 2 bytes for IP (offset) + 2 bytes for CS (segment) of the ISR.

Q46 MODERATE FREQUENTLY ASKED

The PUSH instruction in 8086 decrements SP by:

A. 1
B. 2
C. 4
D. 8

Answer: B. 2
PUSH decrements SP by 2 (word = 2 bytes) before storing. The 8086 stack works with 16-bit words.

Q47 MODERATE FREQUENTLY ASKED

The 8086 stack grows in which direction?

A. Upward (toward higher addresses)
B. Downward (toward lower addresses)
C. Both directions
D. No specific direction

Answer: B. Downward (toward lower addresses)
The 8086 stack grows downward in memory — SP decrements on PUSH and increments on POP.

Q48 EASY

Which 8086 instruction unconditionally transfers control to a subroutine?

A. JMP
B. CALL
C. RET
D. INT

Answer: B. CALL
CALL saves return address on stack and jumps to the subroutine. RET returns from subroutine.

Q49 MODERATE

The HLT instruction in 8086 causes:

A. Processor to reset

B. Processor to enter halt state until interrupt or reset

C. Infinite loop

D. Branch to address 0

Answer: B. Processor to enter halt state until interrupt or reset
HLT stops the processor. It resumes only when an NMI, maskable interrupt (if enabled), or RESET occurs.

Q50 EASY

The 8086 supports word addressing. A word in 8086 is:

A. 4 bits

B. 8 bits

C. 16 bits

D. 32 bits

Answer: C. 16 bits
A word in 8086 = 16 bits (2 bytes). Doubleword (DWORD) = 32 bits.

Q51 MODERATE FREQUENTLY ASKED

What is the maximum offset value within a segment in 8086?

A. 0000H

B. 7FFFH

C. FFFFH

D. FFFFFH

Answer: C. FFFFH
The offset (effective address) is 16-bit, so maximum value = FFFFH (65535 decimal) within a 64 KB segment.

Q52 DIFFICULT

In 8086, XLAT instruction uses which register as a table base?

A. AX
B. BX
C. CX
D. DX

Answer: B. BX
XLAT uses BX as the base address of a 256-byte lookup table and AL as the index; AL ← [DS:BX+AL].

Q53 MODERATE

The LEA instruction in 8086 loads:

A. A byte from memory
B. The effective address (offset) of a memory operand into a register
C. A word from memory
D. The segment address

Answer: B. The effective address (offset) of a memory operand into a register
LEA (Load Effective Address) loads the 16-bit offset (not the data) of a memory operand into a register.

Q54 MODERATE

The TEST instruction in 8086 is similar to AND except:

A. TEST stores the result
B. TEST does not store the result — only sets flags
C. TEST works only with bytes
D. TEST ignores carry flag

Answer: B. TEST does not store the result — only sets flags
TEST performs bitwise AND and sets flags (ZF, SF, PF) but does NOT store the result, unlike AND.

Q55 MODERATE FREQUENTLY ASKED

The LOOP instruction in 8086 decrements ___ and jumps if not zero.

A. AX
B. BX
C. CX
D. DX

Answer: C. CX
LOOP decrements CX; if CX ≠ 0, jumps to specified label. LOOPZ, LOOPNZ also check ZF.

Q56 EASY FREQUENTLY ASKED

Which of the following statement about 8086 is TRUE?

A. 8086 has an 8-bit internal data bus
B. 8086 was introduced in 1982
C. 8086 can address only 64 KB of memory
D. 8086 has a 16-bit internal and external data bus

Answer: D. 8086 has a 16-bit internal and external data bus
8086 is fully 16-bit — both internal and external data buses are 16-bit. It was introduced in 1978 and addresses 1 MB.

Q57 MODERATE FREQUENTLY ASKED

In 8086, the effective address (EA) is computed by:

A. Only the instruction offset
B. Base + Index + Displacement (various combinations
C. Segment × 16
D. IP + CS

Answer: B. Base + Index + Displacement (various combinations)
EA = Base register (BX/BP) + Index register (SI/DI) + Displacement (8 or 16-bit offset), any combination.

Q58 MODERATE FREQUENTLY ASKED

The 8086 MOV AX, [BX+SI+10H] uses which addressing mode?

A. Register
B. Immediate
C. Based Indexed with Displacement
D. Direct

Answer: C. Based Indexed with Displacement
BX (base) + SI (index) + 10H (displacement) = Based Indexed with Displacement addressing mode.

Q59 MODERATE FREQUENTLY ASKED

The 8086 supports which type of I/O?

A. Memory-mapped I/O only
B. Isolated I/O (port-mapped) only
C. Both memory-mapped and isolated I/O
D. Neither

Answer: C. Both memory-mapped and isolated I/O
8086 supports both: memory-mapped I/O (using MOV with memory addresses) and isolated/port I/O (using IN/OUT instructions).

Q60 DIFFICULT FREQUENTLY ASKED

The 8086 PUSH AX instruction stores AX in memory at address:

A. SS:SP
B. SS:SP+2 then decrements SP
C. SS:SP-2 (after decrement)
D. DS:SP

Answer: C. SS:SP-2 (after decrement)
PUSH: SP decrements by 2 first, then AX is written to SS:SP. So stored at SS:(SP-2) effectively.

Q61 MODERATE FREQUENTLY ASKED

The 8086 NEG instruction performs:

A. Logical NOT

B. Two's complement of operand

C. One's complement

D. XOR with FFH

Answer: B. Two's complement of operand
NEG performs two's complement negation: result = 0 – operand. Equivalent to: NOT operand + 1.

Q62 MODERATE

IDIV instruction in 8086 performs:

A. Unsigned division

B. Signed (integer) division

C. BCD division

D. Bit division

Answer: B. Signed (integer) division
IDIV performs signed integer division. DIV is for unsigned division.

Q63 EASY FREQUENTLY ASKED

The 8086 register AH contains:

A. Lower 8 bits of AX

B. Upper 8 bits of AX

C. All 16 bits of AX

D. Upper 4 bits of AX

Answer: B. Upper 8 bits of AX
AX = AH (high byte) + AL (low byte). AH holds bits 8–15 of the 16-bit AX register.

Q64 EASY

The 8086 register AL contains:

A Upper 8 bits of AX

B Lower 8 bits of AX

C All 16 bits of AX

D Lower 4 bits of AX

Answer: B. Lower 8 bits of AX
AL holds bits 0–7 (lower byte) of AX. Similarly, BL, CL, DL are lower bytes of BX, CX, DX.

Q65 MODERATE

The maximum clock speed ever achieved by any standard 8086 variant was:

A 5 MHz

B 8 MHz

C 10 MHz

D 12 MHz

Answer: C. 10 MHz
The fastest standard 8086 variant (8086-1) ran at 10 MHz. IBM PC/XT used 8088 at 4.77 MHz.

Q66 DIFFICULT

The 8086's memory is organized into banks. The even bank contains:

A Data on D8–D15

B Data on D0–D7

C Both D0–D15

D Neither

Answer: B. Data on D0–D7
Even bank (BHE=1, A0=0): bytes at even addresses accessed via D0–D7. Odd bank: via D8–D15.

Q67 EASY

In 8086, the SI register stands for:

A Stack Index

B Source Index

C Segment Index

D System Index

Answer: B. Source Index
SI = Source Index register, typically used as source pointer in string/array operations.

Q68 EASY

In 8086, the DI register stands for:

A Data Index

B Destination Index

C Direct Index

D Double Index

Answer: B. Destination Index
DI = Destination Index register, used as destination pointer in string operations with ES segment.

Q69 MODERATE FREQUENTLY ASKED

The maximum number of simultaneously active segments in 8086 is:

A 1

B 4

C 8

D Unlimited

Answer: B. 4
8086 can have 4 active segments simultaneously: CS (code), DS (data), SS (stack), ES (extra).

Q70 MODERATE FREQUENTLY ASKED

The 8086 string instruction MOVSB moves:

A. A word from DS:SI to ES:DI

B. A byte from DS:SI to ES:DI

C. A byte from ES:DI to DS:SI

D. A block of bytes

Answer: B. A byte from DS:SI to ES:DI
MOVSB moves 1 byte from DS:SI to ES:DI, then updates SI and DI based on Direction Flag.

Q71 MODERATE FREQUENTLY ASKED

In 8086, the REP prefix is used with:

A. Arithmetic instructions

B. String instructions only

C. Jump instructions

D. Interrupt instructions

Answer: B. String instructions only
REP repeats the following string instruction (MOVS, STOS, LODS, etc.) CX times, decrementing CX each time.

Q72 EASY

The 8086 IN instruction is used for:

A. Reading from memory

B. Reading from an I/O port

C. Input of keyboard

D. Initializing registers

Answer: B. Reading from an I/O port
IN AL/AX, port — reads a byte or word from an I/O port into AL or AX register.

Q73 EASY

The 8086 OUT instruction is used for:

A Writing to memory

B Writing to an I/O port

C Output to display

D Setting flags

Answer: B. Writing to an I/O port
OUT port, AL/AX — writes a byte or word from AL or AX to an I/O port.

Q74 MODERATE

Which 8086 instruction exchanges the contents of two registers/memory?

A MOV

B XCHG

C PUSH/POP

D LDS

Answer: B. XCHG
XCHG swaps operands: XCHG AX, BX swaps AX and BX without needing a temporary register.

Q75 MODERATE FREQUENTLY ASKED

The 8086 DAA instruction is used after ADD for:

A Double addition

B Decimal Adjust after Addition for BCD

C Divide and Accumulate

D Data Address Adjustment

Answer: B. Decimal Adjust after Addition for BCD
DAA (Decimal Adjust Accumulator) corrects AL after ADD/ADC to maintain valid BCD representation.

Q76 MODERATE

8086 supports how many addressing modes for data?

A. 3
B. 5
C. 7
D. 9

Answer: C. 7
8086 has 7 main data addressing modes: Register, Immediate, Direct, Register Indirect, Based, Indexed, Based+Indexed.

Q77 DIFFICULT

What is the purpose of the 8086 AAM instruction?

A. ASCII Adjust after Multiplication — converts binary result to BCD
B. Arithmetic Add Memory
C. Auto Address Modify
D. Aligned Address Mode

Answer: A. ASCII Adjust after Multiplication — converts binary result to BCD
AAM (ASCII Adjust after Multiply) adjusts the binary result in AX after a byte multiply to unpacked BCD.

Q78 MODERATE FREQUENTLY ASKED

The 8086 memory address space is organized as:

A. 16-bit flat space
B. 20-bit segmented space (1 MB total)
C. 24-bit flat space
D. Variable size

Answer: B. 20-bit segmented space (1 MB total)
8086 has a 20-bit segmented address space: 1 MB total, organized as overlapping 64 KB segments.

Introduction to 8088

42 questions

Q79 EASY FREQUENTLY ASKED

The Intel 8088 microprocessor was introduced in which year?

A 1976

B 1978

C 1979

D 1982

Answer: C. 1979

Intel 8088 was introduced in 1979, one year after the 8086.

Q80 MODERATE FREQUENTLY ASKED

The 8088 is internally a ___-bit processor.

A 8-bit

B 16-bit

C 32-bit

D 4-bit

Answer: B. 16-bit

8088 has a 16-bit internal architecture (same as 8086) but only an 8-bit external data bus.

Q81 EASY FREQUENTLY ASKED

The external data bus width of 8088 is:

A 4-bit

B 8-bit

C 16-bit

D 32-bit

Answer: B. 8-bit

8088 has an 8-bit external data bus, unlike 8086's 16-bit external bus.

Q82 EASY

The address bus width of 8088 is:

A 16-bit

B 20-bit

C 24-bit

D 32-bit

Answer: B. 20-bit
8088 has a 20-bit address bus, same as 8086, giving 1 MB of addressable memory.

Q83 MODERATE FREQUENTLY ASKED

The instruction prefetch queue of 8088 is:

A 4 bytes

B 6 bytes

C 8 bytes

D 2 bytes

Answer: A. 4 bytes
8088 has a 4-byte instruction queue (vs 8086's 6-byte queue), smaller due to 8-bit bus.

Q84 MODERATE FREQUENTLY ASKED

The 8088 was used in which famous IBM PC?

A IBM PC/AT

B IBM PC 5150 (original IBM PC)

C IBM PS/2 Model 30

D IBM System/360

Answer: B. IBM PC 5150 (original IBM PC)
The original IBM PC (Model 5150, 1981) used the Intel 8088 processor at 4.77 MHz.

Q85 MODERATE FREQUENTLY ASKED

Which is the PRIMARY difference between 8086 and 8088?

A Address bus width

B Internal register size

C External data bus width

D Number of segments

Answer: C. External data bus width
Key difference: 8086 has 16-bit external data bus; 8088 has 8-bit. Internal architecture is identical.

Q86 MODERATE

The 8088's 8-bit external bus made it compatible with which older chips?

A 8085 peripherals

B Z80 chips

C 6800 chips

D Motorola 68000

Answer: A. 8085 peripherals
8088's 8-bit external bus allowed reuse of existing 8085-era 8-bit peripheral chips, reducing system cost.

Q87 MODERATE

The original 8088 operated at which clock frequency?

A 2 MHz

B 5 MHz

C 8 MHz

D 10 MHz

Answer: B. 5 MHz
The original 8088 ran at 5 MHz; the IBM PC used it at 4.77 MHz (3/4 of 6.36 MHz crystal).

Q88 EASY

The number of pins in 8088 is:

A. 28
B. 36
C. 40
D. 44

Answer: C. 40
Intel 8088 is also a 40-pin DIP package, same physical package as 8086.

Q89 MODERATE FREQUENTLY ASKED

In 8088, a 16-bit data transfer requires how many memory cycles?

A. 1
B. 2
C. 4
D. 8

Answer: B. 2
8088's 8-bit bus means a 16-bit word requires TWO memory access cycles instead of one.

Q90 MODERATE

Which of the following is TRUE about 8088?

A. It has 16-bit external data bus
B. It has 24-bit address bus
C. It has 8-bit external data bus and 16-bit internal registers
D. It cannot address 1 MB of memory

Answer: C. It has 8-bit external data bus and 16-bit internal registers
8088: 8-bit external data bus, 16-bit internal registers, 20-bit address bus (1 MB). Correct.

Q91 MODERATE

The IBM PC/XT (1983) also used which processor?

A. 8086
B. 8088
C. 80286
D. 80386

Answer: B. 8088
IBM PC/XT (1983) used the Intel 8088 running at 4.77 MHz, same as original IBM PC.

Q92 MODERATE FREQUENTLY ASKED

The 8088 is best described as:

A. Pure 8-bit processor
B. Pure 16-bit processor
C. 16-bit processor with 8-bit external data bus interface
D. 32-bit processor

Answer: C. 16-bit processor with 8-bit external data bus interface
8088 = 16-bit processor internally with an 8-bit external data bus — a cost-reduced 8086.

Q93 MODERATE

Compared to 8086, the 8088 is generally:

A. Faster in data transfers
B. Slower in data transfers due to 8-bit bus
C. Same speed exactly
D. Faster due to smaller queue

Answer: B. Slower in data transfers due to 8-bit bus
8088 is slower than 8086 for 16-bit transfers (needs 2 cycles vs 1) but was cheaper to implement.

Q94 MODERATE FREQUENTLY ASKED

The 8088 was designed primarily to:

A. Increase processing speed over 8086

B. Reduce system cost by using cheaper 8-bit peripher chips

C. Add Protected Mode support

D. Increase addressable memory

Answer: B. Reduce system cost by using cheaper 8-bit peripheral chips
8088 was created to lower system cost — its 8-bit external bus allowed use of cheaper, widely available 8-bit support chips.

Q95 EASY FREQUENTLY ASKED

The 8088's maximum memory addressing capacity is:

A. 64 KB

B. 256 KB

C. 512 KB

D. 1 MB

Answer: D. 1 MB
8088 has a 20-bit address bus, so it can address 2^20 = 1 MB of memory — same as 8086.

Q96 EASY FREQUENTLY ASKED

The 8088 chip package is:

A. 28-pin DIP

B. 36-pin DIP

C. 40-pin DIP

D. 48-pin DIP

Answer: C. 40-pin DIP
Intel 8088 is housed in a 40-pin DIP (Dual Inline Package), identical in size to the 8086.

Q97 MODERATE

In 8088, how many clock cycles does a minimum bus cycle take?

A. 2
B. 3
C. 4
D. 5

Answer: C. 4
Like 8086, a minimum 8088 bus cycle takes 4 T-states (T1-T4), but 16-bit data needs two such cycles.

Q98 MODERATE FREQUENTLY ASKED

The clock frequency used in the original IBM PC with 8088 was:

A. 2 MHz
B. 4.77 MHz
C. 5 MHz
D. 8 MHz

Answer: B. 4.77 MHz
The original IBM PC (5150, 1981) ran the 8088 at 4.77 MHz — derived as 3/4 of a 14.318 MHz crystal.

Q99 MODERATE

The 8088 microprocessor is manufactured using which technology?

A. TTL
B. CMOS
C. HMOS (NMOS)
D. BiCMOS

Answer: C. HMOS (NMOS)
8088, like 8086, uses Intel's HMOS (High-density N-channel MOS) fabrication technology.

Q100 DIFFICULT

The number of transistors in the 8088 is approximately:

A. 6,500
B. 29,000
C. 50,000
D. 134,000

Answer: B. 29,000
Intel 8088 contains approximately 29,000 transistors — the same as 8086 since the internal architecture is identical.

Q101 DIFFICULT

The 8088 in maximum mode uses which chip for bus control?

A. 8282
B. 8286
C. 8288
D. 8237

Answer: C. 8288
Like 8086 in maximum mode, the 8088 uses the Intel 8288 Bus Controller to generate memory and I/O control signals.

Q102 EASY FREQUENTLY ASKED

The 8088 has how many segment registers?

A. 2
B. 3
C. 4
D. 8

Answer: C. 4
8088 has the same 4 segment registers as 8086: CS (Code), DS (Data), SS (Stack), ES (Extra).

Q103 EASY FREQUENTLY ASKED

The 8088 instruction set is:

A Completely different from 8086

B Identical to 8086

C A subset of 8086

D A superset of 8086

Answer: B. Identical to 8086
8088 has an identical instruction set to 8086 — all 8086 programs run on 8088 without any modification.

Q104 MODERATE FREQUENTLY ASKED

The MN/MX pin on 8088 performs the same function as on 8086, which is:

A Memory/IO select

B Minimum/Maximum mode select

C Multiplexing control

D Mode and Index select

Answer: B. Minimum/Maximum mode select
MN/MX pin: tied HIGH (Vcc) = Minimum Mode (single CPU), tied LOW (GND) = Maximum Mode (multiprocessor). Same as 8086.

Q105 MODERATE FREQUENTLY ASKED

Which of the following is IDENTICAL between 8086 and 8088?

A External data bus width

B Instruction queue size

C Internal register set and architecture

D Number of address pins

Answer: C. Internal register set and architecture
The internal register set, architecture, instruction set, flags, and segment model are identical. Data bus and queue differ.

Q106 MODERATE FREQUENTLY ASKED

The 8088's 4-byte instruction queue is smaller than 8086's 6-byte queue because:

A. 8088 has fewer registers

B. 8088's 8-bit bus fetches 1 byte at a time — a 4-byte queue is sufficient

C. 8088 runs at lower frequency

D. 8088 has smaller ALU

Answer: B. 8088's 8-bit bus fetches 1 byte at a time — a 4-byte queue is sufficient
8088 fetches 1 byte per bus cycle (8-bit bus), so a 4-byte queue avoids waste. 8086 fetches 2 bytes needing 6 bytes.

Q107 MODERATE

The 8088 supports how many I/O port addresses?

A. 256

B. 4096

C. 65536

D. 1048576

Answer: C. 65536
8088 supports 65,536 (64K) I/O port addresses using its 16-bit I/O address space — same as 8086.

Q108 MODERATE FREQUENTLY ASKED

The physical address generation formula in 8088 is:

A. Segment x 8 + Offset

B. Segment x 16 + Offset

C. Segment + Offset

D. Segment x 256 + Offset

Answer: B. Segment x 16 + Offset
8088 uses the same formula as 8086: Physical Address = Segment x 10H (x16) + Offset (Effective Address).

Q109 MODERATE FREQUENTLY ASKED

The 8088 RESET vector points to physical address:

A 00000H

B 0FFFFH

C FFFF0H

D F0000H

Answer: C. FFFF0H
On RESET, both 8086 and 8088 start execution from FFFF0H (CS=FFFFH, IP=0000H). Identical behavior.

Q110 MODERATE FREQUENTLY ASKED

Assertion: 8088 is slower than 8086 for 16-bit data operations. Reason: 8088's 8-bit external data bus requires two bus cycles for a 16-bit transfer.

A Both A and R true; R correctly explains A

B Both A and R true; R does NOT explain A

C A is true; R is false

D A is false; R is true

Answer: A. Both A and R true; R correctly explains A
Both correct and causally linked: 8-bit bus needs 2 cycles per 16-bit word, making it slower than 8086's single cycle.

Q111 EASY

The 8088 general-purpose registers AX, BX, CX, DX are each:

A 8-bit

B 16-bit

C 32-bit

D 4-bit

Answer: B. 16-bit
Like 8086, 8088 has four 16-bit GP registers: AX, BX, CX, DX — each splittable into H and L 8-bit halves.

Q112 DIFFICULT FREQUENTLY ASKED

How does the 8088 handle a 16-bit PUSH instruction?

A. Pushes all 16 bits in one cycle using D0-D15

B. Pushes in two separate 8-bit bus cycles

C. Converts it to an 8-bit push

D. Cannot perform 16-bit PUSH

Answer: B. Pushes in two separate 8-bit bus cycles
Since the bus is only 8-bit, the 8088 performs 16-bit PUSH in two consecutive 8-bit bus cycles — high byte then low byte.

Q113 MODERATE FREQUENTLY ASKED

The 8088's NMI behavior compared to 8086 is:

A. Different — 8088 has no NMI

B. Different vector number

C. Identical — same NMI behavior and vector (Type 2)

D. Different — uses Type 3

Answer: C. Identical — same NMI behavior and vector (Type 2)
8088 and 8086 have identical interrupt architecture including NMI (Type 2) with its vector at 00008H.

Q114 EASY FREQUENTLY ASKED

The 8088 is sometimes called a compromise chip because:

A. It compromises between speed and accuracy

B. It combines 16-bit internal power with 8-bit external bus — balancing cost and performance

C. It runs at a compromised clock frequency

D. It supports only half the instructions of 8086

Answer: B. It combines 16-bit internal power with 8-bit external bus — balancing cost and performance
8088 offers full 16-bit computing power of 8086 internally, but uses a cheaper 8-bit external interface — a cost/performance balance.

Q115 MODERATE FREQUENTLY ASKED

Which CORRECTLY describes the 8088 operating modes?

A Real Mode only

B Protected Mode and Real Mode

C Minimum Mode and Maximum Mode

D Virtual Mode and Real Mode

Answer: C. Minimum Mode and Maximum Mode
Like 8086, 8088 has two operating modes: Minimum Mode (single CPU) and Maximum Mode (multiprocessor). No Protected Mode.

Q116 MODERATE FREQUENTLY ASKED

The 8088 Interrupt Vector Table (IVT) is located at:

A Top of memory (FFFF0H)

B First 1024 bytes (00000H-003FFH)

C Middle of memory

D In the stack segment

Answer: B. First 1024 bytes (00000H-003FFH)
The IVT in 8088 (same as 8086) occupies addresses 00000H to 003FFH — 256 vectors x 4 bytes = 1 KB.

Q117 DIFFICULT

Which Intel processor directly competed with the 8088 from Zilog?

A Z8000

B Z80

C Z8

D Z180

Answer: A. Z8000
Zilog Z8000 was a contemporary 16-bit rival to 8086/8088 in the late 1970s-early 1980s, but Intel's IBM design win made 8088 dominant.

Q118 DIFFICULT

The 8088 SSO pin (S0) in minimum mode indicates:

A. Stack segment overflow

B. Status Signal Output — bus cycle type

C. System Shutdown Output

D. Synchronous Serial Output

Answer: B. Status Signal Output — bus cycle type
SSO (S0) in 8088 minimum mode is a status output replacing 8086's BHE pin, indicating the type of bus cycle in progress.

Q119 EASY FREQUENTLY ASKED

Which operating system ran natively on the 8088-based IBM PC?

A. Unix

B. MS-DOS (PC-DOS)

C. OS/2

D. CP/M-86 only

Answer: B. MS-DOS (PC-DOS)
MS-DOS (PC-DOS 1.0, 1981) was the primary OS for the 8088-based IBM PC, running in Real Mode.

Q120 MODERATE FREQUENTLY ASKED

Which of the following statement about 8088 is FALSE?

A. 8088 has an 8-bit external data bus

B. 8088 was introduced in 1979

C. 8088 has a 6-byte instruction queue

D. 8088 can address 1 MB of memory

Answer: C. 8088 has a 6-byte instruction queue
8088 has a 4-byte instruction queue (NOT 6-byte). 8086 has 6-byte. The other three statements are true.

Architecture of 8086/8088

33 questions

Q121 EASY FREQUENTLY ASKED

The 8086 architecture is divided into how many functional units?

A 1
B 2
C 3
D 4

Answer: B. 2
8086 is divided into 2 units: Bus Interface Unit (BIU) and Execution Unit (EU).

Q122 EASY FREQUENTLY ASKED

Which unit in 8086 is responsible for fetching instructions from memory?

A EU
B BIU
C ALU
D CU

Answer: B. BIU
BIU (Bus Interface Unit) fetches instructions from memory and loads them into the instruction queue.

Q123 EASY FREQUENTLY ASKED

Which unit in 8086 executes instructions?

A BIU
B ALU
C EU
D MMU

Answer: C. EU
EU (Execution Unit) decodes and executes instructions taken from the instruction queue.

Q124 MODERATE FREQUENTLY ASKED

The instruction queue in 8086 follows which data structure principle?

A LIFO

B FIFO

C Random access

D Priority queue

Answer: B. FIFO
The instruction queue is a FIFO (First In, First Out) — instructions are executed in the order they were fetched.

Q125 MODERATE FREQUENTLY ASKED

The overlap of instruction fetch and execution in 8086 is called:

A Multitasking

B Pipelining

C Multiplexing

D Time-sharing

Answer: B. Pipelining
BIU fetching next instruction while EU executes current one is called pipelining — increases throughput.

Q126 EASY

Which register in 8086 holds the offset of the next instruction?

A SP

B BP

C IP

D SI

Answer: C. IP
IP (Instruction Pointer) holds the offset of the NEXT instruction to be fetched from the code segment.

Q127 EASY

The BP register in 8086 stands for:

A Base Pointer

B Bus Pointer

C Block Pointer

D Binary Pointer

Answer: A. Base Pointer
BP = Base Pointer. Typically used to point to the base of the stack frame for parameter access.

Q128 MODERATE FREQUENTLY ASKED

In 8086, the physical address is calculated as:

A Segment + Offset

B Segment × 16 + Offset

C Segment ÷ 16 + Offset

D Segment × 256 + Offset

Answer: B. Segment × 16 + Offset
Physical Address = Segment Register × 16 (or ×10H, shift left 4 bits) + Offset (Effective Address).

Q129 DIFFICULT FREQUENTLY ASKED

If CS = 1000H and IP = 0200H, the physical address is:

A 1200H

B 10200H

C 12000H

D 10020H

Answer: B. 10200H
PA = CS × 10H + IP = 1000H × 10H + 0200H = 10000H + 0200H = 10200H.

Q130 MODERATE FREQUENTLY ASKED

The Trap Flag (TF) in 8086 is used for:

A. Enabling interrupts

B. Single-step mode debugging

C. String operation direction

D. Arithmetic overflow detection

Answer: B. Single-step mode debugging
TF=1 causes 8086 to enter single-step mode — an internal interrupt after each instruction for debugging.

Q131 MODERATE

The Direction Flag (DF) in 8086 controls:

A. Interrupt direction

B. Stack growth

C. String operation direction (increment/decrement)

D. Bus direction

Answer: C. String operation direction (increment/decrement)
DF=0: SI/DI auto-increment after string ops; DF=1: SI/DI auto-decrement (reverse direction).

Q132 MODERATE FREQUENTLY ASKED

The Overflow Flag (OF) is set when:

A. Unsigned overflow occurs

B. Signed arithmetic overflow occurs

C. Result is zero

D. Carry occurs in lower nibble

Answer: B. Signed arithmetic overflow occurs
OF is set when signed arithmetic produces a result too large/small to fit — a signed overflow.

Q133 MODERATE

The Auxiliary Carry Flag (AF) in 8086 is specifically used for:

A. 16-bit carry

B. BCD (Binary Coded Decimal) arithmetic corrections

C. String operations

D. Stack operations

Answer: B. BCD (Binary Coded Decimal) arithmetic corrections
AF is set when carry/borrow occurs from bit 3 to bit 4 — used by DAA/DAS for BCD arithmetic.

Q134 MODERATE

The Sign Flag (SF) in 8086 reflects:

A. MSB of the result

B. LSB of the result

C. Carry from MSB

D. Parity of result

Answer: A. MSB of the result
SF equals the MSB of the result. SF=1 indicates a negative number in signed representation.

Q135 MODERATE

The Parity Flag (PF) in 8086 is set when:

A. Result is odd

B. Lower 8 bits of result have even number of 1s

C. Result is positive

D. Result overflows

Answer: B. Lower 8 bits of result have even number of 1s
PF=1 when the lower byte of the result contains an even number of 1-bits (even parity).

Q136 DIFFICULT

In 8086 maximum mode, which chip generates bus control signals?

A. 8255
B. 8253
C. 8288
D. 8237

Answer: C. 8288
In maximum mode, Intel 8288 Bus Controller chip generates MRDC, MWTC, IORC, IOWC control signals.

Q137 MODERATE FREQUENTLY ASKED

How many total programmer-accessible registers does 8086 have?

A. 8
B. 12
C. 14
D. 16

Answer: C. 14
14 registers: AX,BX,CX,DX (4GP), CS,DS,SS,ES (4segment), SP,BP,SI,DI (4pointer/index), IP, Flags.

Q138 DIFFICULT

The segment:offset addressing in 8086 provides how many bytes of overlap between adjacent segments?

A. None
B. 16 bytes
C. 256 bytes
D. All of 64KB

Answer: B. 16 bytes
Adjacent segments (offset by 1 paragraph = 16 bytes) overlap by 64KB - 16 bytes. Adjacent = 16 bytes different start.

Q139 MODERATE

Which register pair holds the physical starting address of the stack in 8086?

A CS:IP

B DS:SI

C SS:SP

D ES:DI

Answer: C. SS:SP
SS (Stack Segment) holds segment base; SP (Stack Pointer) holds the offset of the stack top.

Q140 DIFFICULT

The 8086 memory model divides 1 MB into how many segments of 16 bytes each?

A 4096 segments

B 65536 paragraphs

C 1024 segments

D 16 segments

Answer: B. 65536 paragraphs
1 MB = 2^{2}■ bytes. Each paragraph = 16 bytes. So 2^{2}■ / 2■ = 2^{1}■ = 65,536 paragraphs.

Q141 MODERATE

In 8086, when BIU finds the instruction queue is full, it:

A Halts execution

B Stops fetching and waits

C Overwrites old instructions

D Sends interrupt

Answer: B. Stops fetching and waits
When the 6-byte instruction queue is full, BIU stops bus fetch cycles until EU consumes instructions.

Q142 MODERATE FREQUENTLY ASKED

Match register with its function:
AX → Accumulator
BX → Base addressing
CX → Loop counter
DX → I/O addressing
Which matching is CORRECT?

A. AX-Accumulator, BX-Counter, CX-Base, DX-I/O

B. AX-Accumulator, BX-Base, CX-Counter, DX-I/O

C. AX-Counter, BX-Accumulator, CX-I/O, DX-Base

D. AX-I/O, BX-Base, CX-Accumulator, DX-Counter

Answer: B. AX-Accumulator, BX-Base, CX-Counter, DX-I/O
AX=Accumulator, BX=Base (addressing), CX=Counter (loops), DX=Data (I/O port addressing). Classic 8086.

Q143 MODERATE FREQUENTLY ASKED

Match segment register to its default use:
CS - Code
DS - Data
SS - Stack
ES - Extra/String destination
Which is CORRECT?

A. CS-Data, DS-Code, SS-Stack, ES-Extra

B. CS-Code, DS-Data, SS-Extra, ES-Stack

C. CS-Code, DS-Data, SS-Stack, ES-Extra/Dest

D. CS-Stack, DS-Data, SS-Code, ES-Extra

Answer: C. CS-Code, DS-Data, SS-Stack, ES-Extra/Dest
CS=Code segment (instructions), DS=Data segment (default data), SS=Stack segment, ES=Extra (string dest).

Q144 MODERATE FREQUENTLY ASKED

Match flag to its type:
CF, ZF, SF, OF, PF, AF → Status flags
TF, IF, DF → Control flags
Which count is correct?

A 4 status, 5 control .
B 6 status, 3 control .
C 5 status, 4 control .
D 3 status, 6 control .

Answer: B. 6 status, 3 control
6 status flags: CF, ZF, SF, OF, PF, AF. 3 control flags: TF (Trap), IF (Interrupt), DF (Direction).

Q145 DIFFICULT

How many segments can fit in the 8086's 1 MB address space simultaneously?

A 4 maximum .
B 16 .
C More than 4 — segments can overlap .
D Exactly 1 MB / 64 KB = 16 non-overlapping .

Answer: C. More than 4 — segments can overlap
Segments can overlap — many more than 16 segment register combinations can map to 1 MB. Only 16 fully non-overlapping 64KB segments possible.

Q146 MODERATE FREQUENTLY ASKED

When the 8086 EU needs data from memory, it requests the bus from:

A ALU .
B BIU .
C Stack .
D IP .

Answer: B. BIU
EU requests the BIU to perform memory read/write operations — EU itself has no direct bus access.

Q147 DIFFICULT

The 8086 queue status (QS0, QS1) signals in maximum mode indicate:

A. Memory queue size
B. Status of instruction prefetch queue
C. DMA queue
D. Interrupt queue

Answer: B. Status of instruction prefetch queue
QS0 and QS1 outputs in max mode tell external co-processors about queue status: first byte, continuation, empty, etc.

Q148 MODERATE FREQUENTLY ASKED

In 8086, the instruction queue improves performance by:

A. Reducing memory size needed
B. Overlapping fetch and execute (pipelining)
C. Increasing clock speed
D. Reducing number of registers

Answer: B. Overlapping fetch and execute (pipelining)
The instruction queue enables pipelining: BIU pre-fetches ahead so EU rarely waits for next instruction.

Q149 MODERATE FREQUENTLY ASKED

When EU encounters a branch (JUMP) in 8086, the instruction queue is:

A. Saved for later
B. Flushed and refilled from new address
C. Paused
D. Moved to stack

Answer: B. Flushed and refilled from new address
On a taken branch (JMP/CALL/interrupt), the queue is flushed — prefetched bytes are invalid. BIU refills from new address.

Q150 DIFFICULT

The 8086 BIU fetches a new byte from memory whenever the queue has at least ___ empty byte(s).

A 1

B 2

C 3

D 4

Answer: B. 2
8086 BIU initiates a bus cycle to fetch 2 bytes whenever there are 2 or more empty bytes in the 6-byte queue.

Q151 MODERATE FREQUENTLY ASKED

In 8086, the effective address for [BP+DI+20H] is computed as:

A BP+DI

B BP+DI+20H

C CS:BP+DI+20H

D SS:BP+DI+20H

Answer: B. BP+DI+20H
EA = BP + DI + 20H. Note: when BP is used, the default segment is SS (stack segment), not DS.

Q152 EASY FREQUENTLY ASKED

Which register in 8086 is automatically used with PUSH/POP and CALL/RET?

A BP

B IP

C SP

D SI

Answer: C. SP
SP (Stack Pointer) is automatically incremented/decremented by PUSH, POP, CALL, RET instructions.

Q153 MODERATE FREQUENTLY ASKED

The 8086 internal ALU size is:

A 8-bit

B 16-bit

C 32-bit

D 4-bit

Answer: B. 16-bit
The 8086 ALU (in EU) is 16-bit wide, matching the 16-bit register width of the processor.

Bus System
38 questions

Q154 EASY FREQUENTLY ASKED

A bus in computer architecture is:

A. A single wire
B. A group of parallel wires for signal transfer
C. A control signal only
D. A memory component

Answer: B. A group of parallel wires for signal transfer
A bus is a set of parallel conductors (wires/PCB traces) that carry signals between components.

Q155 EASY FREQUENTLY ASKED

The data bus in 8086 is:

A. Unidirectional
B. Bidirectional
C. Multiplexed only
D. Neither directional

Answer: B. Bidirectional
The data bus is bidirectional — data flows TO memory (write) and FROM memory (read).

Q156 EASY FREQUENTLY ASKED

The address bus in 8086 is:

A. Bidirectional
B. Unidirectional
C. Always multiplexed
D. Tristate only

Answer: B. Unidirectional
The address bus is unidirectional — addresses flow only FROM the CPU to memory/I/O devices.

Q157 MODERATE FREQUENTLY ASKED

In 8086, the lower 16 lines AD0–AD15 are:

A. Dedicated address lines

B. Dedicated data lines

C. Multiplexed address/data lines

D. Control lines

Answer: C. Multiplexed address/data lines
AD0–AD15 are multiplexed: carry ADDRESS during T1 clock cycle, then DATA during T2, T3, T4.

Q158 MODERATE

The upper 4 address lines A16–A19 in 8086 carry:

A. Only address

B. Only data

C. Address (in T1) and status S3–S6 (in T2–T4)

D. Only control signals

Answer: C. Address (in T1) and status S3–S6 (in T2–T4)
A16/S3–A19/S6 lines carry address bits during T1 and status bits S3–S6 during other clock states.

Q159 MODERATE FREQUENTLY ASKED

The control bus is which type in terms of direction?

A. Always bidirectional

B. Always unidirectional

C. Both — some signals are input to CPU, some output

D. Neither

Answer: C. Both — some signals are input to CPU, some output
Control bus has mixed-direction lines: RD/WR are outputs; READY/INTR/NMI are inputs to CPU.

Q160 MODERATE FREQUENTLY ASKED

The READY signal in 8086 is used for:

A. DMA transfer

B. Interrupt acknowledge

C. Inserting wait states for slow memory/IO

D. Bus request by DMA

Answer: C. Inserting wait states for slow memory/IO
If READY=0 (low), 8086 inserts wait states (T-wait cycles) to accommodate slow memory/peripherals.

Q161 MODERATE FREQUENTLY ASKED

The RESET signal causes 8086 to start execution from physical address:

A. 00000H

B. FFFF0H

C. F0000H

D. 0FFFFH

Answer: B. FFFF0H
On RESET: CS=FFFFH, IP=0000H → PA = FFFFH×10H + 0000H = FFFF0H. First instruction at FFFF0H.

Q162 MODERATE

The INTR signal in 8086 is:

A. Non-maskable interrupt

B. Maskable hardware interrupt

C. Software interrupt

D. Bus interrupt request

Answer: B. Maskable hardware interrupt
INTR is the maskable hardware interrupt — can be enabled/disabled with STI/CLI (IF flag).

Q163 EASY FREQUENTLY ASKED

The NMI signal in 8086 stands for:

A Normal Mode Interrupt

B Non-Maskable Interrupt

C Negative Memory Interrupt

D Null Memory Interface

Answer: B. Non-Maskable Interrupt
NMI = Non-Maskable Interrupt. Cannot be disabled by software; has higher priority than INTR.

Q164 MODERATE

The HOLD/HLDA signals in 8086 are used for:

A Halting CPU

B DMA bus request/acknowledge

C Interrupt service

D Memory hold

Answer: B. DMA bus request/acknowledge
HOLD: DMA requests bus. HLDA (Hold Acknowledge): CPU acknowledges, releases bus to DMA controller.

Q165 MODERATE FREQUENTLY ASKED

The M/IO signal in 8086 (minimum mode) indicates:

A Memory versus I/O operation

B Minimum versus Maximum mode

C Multiplexed versus I/O

D Modular input/output

Answer: A. Memory versus I/O operation
M/IO=1: memory reference; M/IO=0: I/O port reference. Distinguishes memory and I/O operations.

Q166 MODERATE FREQUENTLY ASKED

The BHE signal in 8086 stands for:

A. Bus High Enable
B. Byte High Enable
C. Block Hold Enable
D. Bus Hold Enable

Answer: B. Byte High Enable
BHE (Byte High Enable, active low) enables the upper byte (D8–D15) of the 16-bit data bus.

Q167 EASY FREQUENTLY ASKED

The RD signal in 8086 (active low) indicates:

A. Write operation
B. Read operation in progress
C. Reset
D. DMA request

Answer: B. Read operation in progress
RD (active low) = CPU is performing a memory or I/O read. The addressed device should drive data onto bus.

Q168 EASY FREQUENTLY ASKED

The WR signal in 8086 (active low) indicates:

A. Write operation in progress
B. Read operation
C. Wait state
D. Reset operation

Answer: A. Write operation in progress
WR (active low) = CPU is performing a write. Valid data is on the data bus for memory/IO to capture.

Q169 EASY FREQUENTLY ASKED

How many address lines are needed to address 1 MB of memory?

A 16

B 18

C 20

D 24

Answer: C. 20

2^{20} = 1,048,576 bytes = 1 MB. Therefore exactly 20 address lines are required.

Q170 MODERATE FREQUENTLY ASKED

A 24-bit address bus can address a maximum of:

A 4 MB

B 8 MB

C 16 MB

D 32 MB

Answer: C. 16 MB

2^{24} = 16,777,216 bytes = 16 MB. This is the capacity of the 80286's address bus.

Q171 MODERATE FREQUENTLY ASKED

A 32-bit address bus can address a maximum of:

A 256 MB

B 1 GB

C 4 GB

D 16 GB

Answer: C. 4 GB

2^{32} = 4,294,967,296 bytes = 4 GB. This is the capacity of 80386 and Pentium address buses.

Q172 DIFFICULT

The purpose of the address latch (e.g., 74LS373) in 8086 systems is:

A. To amplify signals

B. To hold/demultiplex addresses from AD0–AD15

C. To control memory speed

D. To generate interrupts

Answer: B. To hold/demultiplex addresses from AD0–AD15
Latches like 74LS373/8282 capture and hold the address presented by AD0–AD15 during ALE pulse, demultiplexing it.

Q173 DIFFICULT

During which clock state does 8086 put the address on the bus?

A. T2

B. T3

C. T1

D. T4

Answer: C. T1
In T1 state, 8086 puts the address on AD0–AD19 and asserts ALE to latch it externally.

Q174 MODERATE FREQUENTLY ASKED

How many bytes can 8086 transfer in one bus cycle with 16-bit data bus?

A. 1 byte

B. 2 bytes

C. 4 bytes

D. 8 bytes

Answer: B. 2 bytes
8086 transfers 2 bytes (1 word = 16 bits) per bus cycle using its 16-bit data bus.

Q175 MODERATE

The 8086 bus cycle consists of minimum how many T-states?

A. 2
B. 3
C. 4
D. 5

Answer: C. 4
A minimum 8086 bus cycle has 4 T-states: T1 (address), T2 (data setup), T3 (data valid), T4 (complete).

Q176 DIFFICULT

Wait states (T-wait) in 8086 are inserted between which T-states?

A. T1 and T2
B. T2 and T3
C. T3 and T4
D. T4 and T1

Answer: C. T3 and T4
Wait states are inserted between T3 and T4 when READY signal is low, extending the bus cycle.

Q177 DIFFICULT

The 8086 bus operates in which mode for idle periods?

A. T-idle
B. T-hold
C. Ti (idle state)
D. T-wait

Answer: C. Ti (idle state)
When no bus cycle is needed, 8086 enters Ti (idle) states — NOP cycles on the bus.

Q178 DIFFICULT

The S0, S1, S2 status signals in 8086 maximum mode indicate:

A. Interrupt type

B. Type of bus cycle being performed

C. Memory size

D. DMA type

Answer: B. Type of bus cycle being performed
S0–S2 outputs in maximum mode encode the type of bus cycle: memory read/write, I/O, interrupt ack, halt, etc.

Q179 MODERATE

Which signal in 8086 indicates that a bus cycle is in progress?

A. ALE

B. BHE

C. LOCK

D. INTA

Answer: A. ALE
ALE (Address Latch Enable) pulse marks the start of a bus cycle — address is valid on the bus.

Q180 DIFFICULT

The LOCK prefix in 8086 instruction set is used to:

A. Lock the keyboard

B. Assert LOCK signal to prevent other bus masters during critical section

C. Lock the interrupt

D. Lock the DMA

Answer: B. Assert LOCK signal to prevent other bus masters during critical section
LOCK prefix asserts the LOCK pin, preventing other bus masters (in max mode) from acquiring the bus.

Q181 MODERATE FREQUENTLY ASKED

The INTA signal in 8086 is generated during:

A. Reset
B. Interrupt acknowledge cycle
C. DMA transfer
D. Memory read

Answer: B. Interrupt acknowledge cycle
INTA (Interrupt Acknowledge, active low) is driven low during two consecutive bus cycles to acknowledge an interrupt.

Q182 MODERATE FREQUENTLY ASKED

Match bus type to its direction:
Data Bus → Bidirectional
Address Bus → Unidirectional (CPU out)
Control Bus → Mixed (bidirectional depending on signal)
Which is CORRECTLY described?

A. Data=Unidirectional, Address=Bidirectional, Control=Mixed
B. Data=Bidirectional, Address=Unidirectional, Control=Mixed
C. Data=Mixed, Address=Unidirectional, Control=Bidirectional
D. All three are bidirectional

Answer: B. Data=Bidirectional, Address=Unidirectional, Control=Mixed
Data bus: bidirectional (read/write). Address bus: unidirectional (CPU to memory). Control: mixed (RD/WR out, READY/INTR in).

Q183 MODERATE FREQUENTLY ASKED

If DS = 3000H and effective address = 0500H, the physical address is:

A. 3500H
B. 30500H
C. 35000H
D. 30050H

Answer: B. 30500H
PA = DS × 10H + EA = 3000H × 10H + 0500H = 30000H + 0500H = 30500H.

Q184 MODERATE FREQUENTLY ASKED

If SS = 2000H and SP = FF00H, the physical address of stack top is:

A. 2FF00H
B. 20F00H
C. 22F00H
D. 2F000H

Answer: A. 2FF00H
PA = SS × 10H + SP = 2000H × 10H + FF00H = 20000H + FF00H = 2FF00H.

Q185 DIFFICULT

Two segment registers S1=1000H and S2=1400H start segments that:

A. Do not overlap
B. Partially overlap — second starts inside first
C. Are identical
D. S2 is entirely outside S1

Answer: B. Partially overlap — second starts inside first
S1 covers 10000H–1FFFFH; S2 covers 14000H–23FFFH. S2 starts at 14000H which is inside S1's range → partial overlap.

Q186 EASY FREQUENTLY ASKED

Which of the following is a function of the control bus?

A. Carries data between CPU and memory
B. Specifies the memory or I/O address
C. Carries read/write, interrupt, clock and synchronization signals
D. Provides power to memory chips

Answer: C. Carries read/write, interrupt, clock and synchronization signals
Control bus carries timing, read/write control, interrupt, reset, bus-request/grant and other coordination signals.

Q187 MODERATE FREQUENTLY ASKED

The 8086 bus cycle in minimum mode uses which signals as control?

A S0-S2

B RD, WR, M/IO, ALE, DT/R, DEN

C LOCK, QS0, QS1

D MRDC, MWTC

Answer: B. RD, WR, M/IO, ALE, DT/R, DEN
Minimum mode: 8086 directly generates RD, WR, M/IO, ALE, DT/R (data transmit/receive), DEN (data enable).

Q188 MODERATE FREQUENTLY ASKED

The DEN signal in 8086 minimum mode stands for:

A Data Enable — enables data bus transceivers

B Decoder Enable

C Direct Entry Node

D Data End signal

Answer: A. Data Enable — enables data bus transceivers
DEN (Data Enable) enables the external 8286 data bus transceivers for data transfer.

Q189 MODERATE FREQUENTLY ASKED

The DT/R signal in 8086 minimum mode indicates:

A Data transfer rate

B Data Transmit (1) or Data Receive (0)

C DMA transfer request

D Data test result

Answer: B. Data Transmit (1) or Data Receive (0)
DT/R: when HIGH = transmit (CPU sending data); when LOW = receive (CPU receiving data). Controls 8286 direction.

Q190 MODERATE FREQUENTLY ASKED

The 8086 has a multiplexed address/data bus. The number of multiplexed lines is:

A. 8
B. 16
C. 20
D. 24

Answer: B. 16
8086 has 16 multiplexed AD0–AD15 lines (address/data) plus 4 A16–A19/S3–S6 lines.

Q191 EASY FREQUENTLY ASKED

A 16-bit address bus can address how many memory locations?

A. 256
B. 4096
C. 32768
D. 65536

Answer: D. 65536
2^{16} = 65,536 memory locations. That's why 8-bit CPUs with 16-bit address bus could address 64 KB.

80286 Microprocessor

30 questions

Q192 EASY FREQUENTLY ASKED

The Intel 80286 was introduced in which year?

A 1978

B 1980

C 1982

D 1984

Answer: C. 1982

Intel 80286 was introduced in February 1982.

Q193 EASY FREQUENTLY ASKED

The 80286 is a ___-bit microprocessor.

A 8-bit

B 16-bit

C 32-bit

D 64-bit

Answer: B. 16-bit

80286 is a 16-bit processor — both internal registers and external data bus are 16-bit.

Q194 EASY FREQUENTLY ASKED

The address bus width of 80286 is:

A 20-bit

B 22-bit

C 24-bit

D 32-bit

Answer: C. 24-bit

80286 has a 24-bit address bus, allowing it to address 2^{24} = 16 MB of physical memory.

Q195 EASY FREQUENTLY ASKED

What is the maximum physical memory addressable by 80286?

A 1 MB

B 4 MB

C 16 MB

D 64 MB

Answer: C. 16 MB
80286 with 24-bit address bus: 2²■ = 16,777,216 bytes = 16 MB.

Q196 EASY FREQUENTLY ASKED

The 80286 operates in how many modes?

A 1

B 2

C 3

D 4

Answer: B. 2
80286 has two operating modes: Real Address Mode (Real Mode) and Protected Virtual Address Mode.

Q197 EASY FREQUENTLY ASKED

Which mode of 80286 is compatible with 8086 programs?

A Protected Mode

B Virtual Mode

C Real Mode

D System Mode

Answer: C. Real Mode
Real Mode in 80286 behaves like a fast 8086, running 8086 programs without modification.

Q198 MODERATE

In Protected Mode, 80286 can use virtual memory up to:

A. 1 MB
B. 16 MB physical
C. 1 GB virtual
D. 4 GB virtual

Answer: C. 1 GB virtual
80286 Protected Mode: 16 MB physical memory + up to 1 GB virtual address space per task.

Q199 MODERATE FREQUENTLY ASKED

The IBM PC/AT (1984) used which microprocessor?

A. 8086
B. 8088
C. 80286
D. 80386

Answer: C. 80286
IBM PC/AT (Advanced Technology, 1984) was the first mainstream PC based on Intel 80286.

Q200 MODERATE

The 80286 has how many pins?

A. 40
B. 48
C. 68
D. 100

Answer: C. 68
80286 has 68 pins in a PGA (Pin Grid Array) package, compared to 8086's 40 pins.

Q201 MODERATE

The clock speed of the original IBM PC/AT with 80286 was:

A. 5 MHz
B. 6 MHz
C. 8 MHz
D. 10 MHz

Answer: C. 8 MHz
IBM PC/AT used 80286 at 8 MHz. The chip itself ran at 8, 10, 12, and 16 MHz variants.

Q202 MODERATE FREQUENTLY ASKED

Protected Mode in 80286 provides which primary feature?

A. Faster execution
B. Memory protection with privilege levels
C. Floating-point support
D. Built-in graphics

Answer: B. Memory protection with privilege levels
Protected Mode provides hardware-enforced memory protection, privilege levels (rings 0–3), and virtual memory.

Q203 MODERATE FREQUENTLY ASKED

80286 Protected Mode has how many privilege rings?

A. 2
B. 4
C. 8
D. 16

Answer: B. 4
80286 has 4 privilege levels: Ring 0 (OS kernel, most privileged) to Ring 3 (user application, least).

Q204 DIFFICULT FREQUENTLY ASKED

A major limitation of 80286 is that from Protected Mode, it CANNOT easily switch to:

A. Virtual Mode

B. Real Mode without hardware reset

C. Supervisor Mode

D. Extended Mode

Answer: B. Real Mode without hardware reset
80286 cannot switch from Protected Mode back to Real Mode without a CPU reset — fixed in 80386.

Q205 DIFFICULT

The 80286 introduced which memory management tables?

A. GDT and LDT only

B. PDT (Page Descriptor Table)

C. GDT, LDT, and IDT

D. TLB only

Answer: A. GDT and LDT only
80286 introduced GDT (Global Descriptor Table), LDT (Local Descriptor Table), and IDT in Protected Mode.

Q206 DIFFICULT

The 80286 contains approximately how many transistors?

A. 29,000

B. 134,000

C. 275,000

D. 1,200,000

Answer: B. 134,000
Intel 80286 contains approximately 134,000 transistors (vs 8086's ~29,000).

Q207 MODERATE FREQUENTLY ASKED

The improvement of 80286 over 8086 in memory capacity is:

A. Same — both 1 MB

B. 80286 = 16× more (16 MB vs 1 MB)

C. 80286 = 4× more (4 MB vs 1 MB)

D. 80286 = 256× more

Answer: B. 80286 = 16× more (16 MB vs 1 MB)
8086 = 1 MB (20-bit bus); 80286 = 16 MB (24-bit bus). 16 MB / 1 MB = 16× improvement.

Q208 DIFFICULT

The 80286 GDTR (Global Descriptor Table Register) stores:

A. Physical address and size of GDT

B. Virtual address of GDT

C. LDT location

D. IDT location

Answer: A. Physical address and size of GDT
GDTR holds the 24-bit linear base address and 16-bit limit of the Global Descriptor Table.

Q209 DIFFICULT

A segment descriptor in 80286 is how many bytes wide?

A. 2 bytes

B. 4 bytes

C. 8 bytes

D. 16 bytes

Answer: C. 8 bytes
Each segment descriptor in the 80286 GDT/LDT is 8 bytes (64 bits) wide.

Q210 DIFFICULT

The 80286 maximum virtual memory per task in Protected Mode is:

A. 16 MB
B. 64 MB
C. 1 GB
D. 4 GB

Answer: C. 1 GB
80286 Protected Mode provides up to 1 GB of virtual address space per task using segmentation.

Q211 EASY

The 80286 data bus is how many bits wide?

A. 8-bit
B. 16-bit
C. 24-bit
D. 32-bit

Answer: B. 16-bit
80286 has a full 16-bit external data bus — same width as 8086 but with much more memory.

Q212 DIFFICULT

The 80286 was used in which Microsoft OS that exploited Protected Mode?

A. MS-DOS 3.0
B. OS/2 1.x
C. Windows 1.0
D. CP/M-86

Answer: B. OS/2 1.x
IBM/Microsoft OS/2 1.x (1987) was the first OS to use 80286 Protected Mode for multi-tasking.

Q213 MODERATE FREQUENTLY ASKED

The 80286 has how many privilege levels (rings)?

A. 2
B. 3
C. 4
D. 8

Answer: C. 4
80286 implements 4 privilege levels: Ring 0 (kernel), Ring 1, Ring 2, Ring 3 (user applications).

Q214 MODERATE FREQUENTLY ASKED

In 80286 Real Mode, memory access is limited to:

A. 64 KB
B. 1 MB
C. 16 MB
D. 4 GB

Answer: B. 1 MB
In Real Mode, 80286 behaves like a fast 8086 — limited to 1 MB (20-bit effective addressing).

Q215 MODERATE FREQUENTLY ASKED

Which of the following about 80286 is FALSE?

A. 80286 introduced Protected Mode
B. 80286 has 24-bit address bus
C. 80286 can freely switch from Protected to Real Mode
D. 80286 has 4 privilege levels

Answer: C. 80286 can freely switch from Protected to Real Mode
80286 CANNOT easily switch from Protected to Real Mode without a hardware reset. This is FALSE/limitation.

Q216 DIFFICULT FREQUENTLY ASKED

The 80286's segment descriptor contains which information?

A Segment base address only

B Base address, limit, access rights, and type

C Only limit

D Only access rights

Answer: B. Base address, limit, access rights, and type
An 80286 segment descriptor (8 bytes) contains: 24-bit base, 16-bit limit, access rights byte, and attribute bits.

Q217 MODERATE FREQUENTLY ASKED

80286 Real Mode segment limit is always:

A Variable (programmable)

B Fixed at 64 KB

C Fixed at 16 KB

D Fixed at 1 MB

Answer: B. Fixed at 64 KB
In 80286 Real Mode (like 8086), all segment sizes are fixed at maximum 64 KB (FFFFH offset).

Q218 MODERATE FREQUENTLY ASKED

In 80286 Protected Mode, segments can be how large?

A 64 KB maximum

B 16 MB maximum (24-bit base)

C Variable 1 byte to 64 KB

D Fixed 4 KB

Answer: B. 16 MB maximum (24-bit base)
In Protected Mode, 80286 segments: base is 24-bit (16 MB range) with 16-bit limit up to 64 KB per segment.

Q219 DIFFICULT

The 80286 Local Descriptor Table (LDT) is for:

A. Global OS-level segment descriptions

B. Per-task private segment descriptions

C. Interrupt handling

D. DMA control

Answer: B. Per-task private segment descriptions
LDT holds per-task segment descriptors (private to each task), while GDT holds global/system-wide descriptors.

Q220 DIFFICULT

80286 SGDT instruction is used to:

A. Set up the GDT

B. Store GDTR (save the GDT base+limit)

C. Switch to Protected Mode

D. Load segment descriptor

Answer: B. Store GDTR (save the GDT base+limit)
SGDT (Store GDT Register) saves the GDTR contents (base address + limit) to a memory location.

Q221 DIFFICULT FREQUENTLY ASKED

The 80286 CALLgate provides:

A. A way to call OS routines at higher privilege

B. A hardware interrupt

C. A DMA channel

D. A memory bank switch

Answer: A. A way to call OS routines at higher privilege
A call gate in 80286 Protected Mode allows controlled privilege escalation — user code calling OS at ring 0.

80386 Microprocessor

31 questions

Q222 EASY FREQUENTLY ASKED

The Intel 80386 was introduced in which year?

A 1984
B 1985
C 1986
D 1988

Answer: B. 1985
Intel 80386 (i386) was introduced in October 1985.

Q223 EASY FREQUENTLY ASKED

The 80386 is Intel's first ___-bit microprocessor.

A 16-bit
B 24-bit
C 32-bit
D 64-bit

Answer: C. 32-bit
80386 was Intel's first true 32-bit processor with 32-bit registers, 32-bit data bus, and 32-bit address bus.

Q224 EASY FREQUENTLY ASKED

The data bus width of 80386DX is:

A 16-bit
B 24-bit
C 32-bit
D 64-bit

Answer: C. 32-bit
80386DX has a full 32-bit external data bus, enabling 32 bits transferred in one bus cycle.

Q225 EASY FREQUENTLY ASKED

The address bus width of 80386 is:

A 24-bit

B 32-bit

C 36-bit

D 40-bit

Answer: B. 32-bit
80386 has a 32-bit address bus, enabling addressing of 2^{32} = 4 GB of physical memory.

Q226 EASY FREQUENTLY ASKED

Maximum physical memory addressable by 80386 is:

A 16 MB

B 64 MB

C 256 MB

D 4 GB

Answer: D. 4 GB
80386 with 32-bit address bus: 2^{32} = 4,294,967,296 bytes = 4 GB.

Q227 MODERATE FREQUENTLY ASKED

The 80386 operates in how many modes?

A 2

B 3

C 4

D 5

Answer: B. 3
80386 has 3 operating modes: Real Mode, Protected Mode, and Virtual 8086 (V86) Mode.

Q228 MODERATE FREQUENTLY ASKED

Virtual 8086 mode in 80386 allows:

A. Running 16-bit protected apps

B. Running multiple 8086 programs simultaneously under a multitasking OS

C. Accessing 4 GB directly from each app

D. Floating-point operations only

Answer: B. Running multiple 8086 programs simultaneously under a multitasking OS
V86 Mode lets 80386 run multiple 8086 programs simultaneously, each in protected virtual memory space.

Q229 MODERATE FREQUENTLY ASKED

80386 was the first Intel x86 processor to support:

A. Protected Mode

B. Paging (hardware demand-paged virtual memory)

C. Segmentation

D. Hardware interrupts

Answer: B. Paging (hardware demand-paged virtual memory)
80386 was the first x86 chip to include a hardware Paging Unit for demand-paged virtual memory.

Q230 MODERATE FREQUENTLY ASKED

The page size in 80386 paging is:

A. 512 bytes

B. 1 KB

C. 4 KB

D. 64 KB

Answer: C. 4 KB
80386 paging uses 4 KB (4096-byte) pages as the basic unit of virtual-to-physical memory translation.

Q231 MODERATE FREQUENTLY ASKED

Unlike 80286, the 80386 can switch between Protected and Real modes:

A Not at all

B Only with hardware reset

C Freely in software without reset

D Only with OS assistance

Answer: C. Freely in software without reset
80386 fixed the 80286 limitation — it can freely switch between modes in software using CR0 register.

Q232 MODERATE FREQUENTLY ASKED

The 80386DX has which external bus configuration?

A 32-bit data, 32-bit address

B 32-bit data, 24-bit address

C 16-bit data, 32-bit address

D 16-bit data, 24-bit address

Answer: A. 32-bit data, 32-bit address
80386DX (full version): 32-bit external data bus + 32-bit external address bus. Full 32-bit chip.

Q233 DIFFICULT

The 80386SX has which external bus configuration?

A 32-bit data, 32-bit address

B 32-bit data, 24-bit address

C 16-bit data, 24-bit address

D 16-bit data, 32-bit address

Answer: C. 16-bit data, 24-bit address
80386SX: 32-bit internal architecture but 16-bit external data bus and 24-bit external address bus.

Q234 DIFFICULT

80386 contains approximately how many transistors?

A. 134,000
B. 275,000
C. 1,200,000
D. 3,100,000

Answer: B. 275,000
Intel 80386 contains approximately 275,000 transistors — twice as many as 80286.

Q235 MODERATE

The 80386 clock speed range was approximately:

A. 5–10 MHz
B. 12–40 MHz
C. 50–100 MHz
D. 100–200 MHz

Answer: B. 12–40 MHz
80386 variants ranged from 12 MHz (80386-12) to 40 MHz (80386-40).

Q236 DIFFICULT

The 80386DX has how many pins?

A. 40
B. 68
C. 100
D. 132

Answer: D. 132
80386DX uses a 132-pin PGA (Pin Grid Array) package.

Q237 MODERATE FREQUENTLY ASKED

The improvement of 80386 over 80286 in memory capacity is:

A. Same
B. 80386 = 256× more (4 GB vs 16 MB)
C. 80386 = 16× more
D. 80386 = 64× more

Answer: B. 80386 = 256× more (4 GB vs 16 MB)
80286 = 16 MB; 80386 = 4 GB. 4 GB / 16 MB = 256× improvement in physical memory addressing.

Q238 DIFFICULT FREQUENTLY ASKED

The 80386 Control Register CR0 bit 0 (PE) controls:

A. Paging enable
B. Protection Enable (Real vs Protected mode)
C. Cache enable
D. Coprocessor enable

Answer: B. Protection Enable (Real vs Protected mode)
CR0.PE=0: Real Mode; CR0.PE=1: Protected Mode. Setting PE switches the 80386 to Protected Mode.

Q239 DIFFICULT

The 80386 Control Register CR0 bit 31 (PG) controls:

A. Protection
B. Paging unit enable/disable
C. Cache
D. Floating point

Answer: B. Paging unit enable/disable
CR0.PG=1 enables the paging unit. CR0.PG=0 disables paging (flat physical memory used).

Q240 DIFFICULT

The 80386 page directory has how many entries?

A. 256
B. 512
C. 1024
D. 4096

Answer: C. 1024
80386 two-level paging: Page Directory has 1024 (2¹■) entries, each pointing to a Page Table.

Q241 EASY FREQUENTLY ASKED

The 80386 extended general-purpose registers are named:

A. AX, BX, CX, DX
B. EAX, EBX, ECX, EDX
C. RAX, RBX, RCX, RDX
D. MAX, MBX, MCX, MDX

Answer: B. EAX, EBX, ECX, EDX
80386 extended 16-bit registers to 32-bit by adding 'E' prefix: EAX, EBX, ECX, EDX.

Q242 DIFFICULT

The 80386 has how many debug registers?

A. 2
B. 4
C. 6
D. 8

Answer: D. 8
80386 has 8 debug registers (DR0–DR7): DR0–DR3 for breakpoint addresses, DR6 for status, DR7 for control.

Q243 MODERATE FREQUENTLY ASKED

80386 Linear (virtual) address space is:

A. 1 MB
B. 16 MB
C. 4 GB
D. 1 TB

Answer: C. 4 GB
80386 provides a 32-bit linear address space = 4 GB per task (with paging, this is virtual/linear space).

Q244 MODERATE

Which 80386 mode allows protected multitasking AND backward compatibility with 8086 apps?

A. Real Mode
B. Protected Mode without paging
C. Protected Mode with Virtual 8086 Mode
D. System Management Mode

Answer: C. Protected Mode with Virtual 8086 Mode
Protected Mode + Virtual 8086 Mode: each V86 task runs an 8086 program in its own protected 1 MB space.

Q245 MODERATE FREQUENTLY ASKED

Which of the following about 80386 is TRUE?

A. 80386 introduced Protected Mode in x86
B. 80386 has a 24-bit address bus
C. 80386 was the first Intel 32-bit x86 CPU
D. 80386 has no paging support

Answer: C. 80386 was the first Intel 32-bit x86 CPU
80386 was indeed the first 32-bit Intel x86 CPU (1985). It introduced paging. Protected Mode was 80286.

Q246 DIFFICULT FREQUENTLY ASKED

In 80386 paging, a 32-bit linear address is split as:

A. 10+10+12 bits (dir+table+offset)

B. 12+10+10 bits

C. 16+16 bits

D. 20+12 bits

Answer: A. 10+10+12 bits (dir+table+offset)
80386: bits 31–22 = Page Directory index (10 bits), bits 21–12 = Page Table index (10 bits), bits 11–0 = offset (12 bits).

Q247 MODERATE FREQUENTLY ASKED

The 80386's EIP register corresponds to 8086's:

A. AX

B. IP (Instruction Pointer), extended to 32 bits

C. CS

D. SP

Answer: B. IP (Instruction Pointer), extended to 32 bits
EIP is the 32-bit extended version of the 8086's 16-bit IP (Instruction Pointer) register.

Q248 MODERATE FREQUENTLY ASKED

In 80386, EFLAGS extends the 8086 FLAGS register to how many bits?

A. 16-bit

B. 32-bit

C. 64-bit

D. 24-bit

Answer: B. 32-bit
80386 EFLAGS is 32 bits wide (extending the 8086 16-bit FLAGS), adding bits for VM, RF, NT, IOPL flags.

Q249 DIFFICULT FREQUENTLY ASKED

The VM bit in 80386 EFLAGS register, when set to 1, indicates:

A. Video Mode

B. Virtual 8086 Mode is active

C. Virtual Memory enabled

D. Version Mode

Answer: B. Virtual 8086 Mode is active
VM (Virtual Mode) bit in EFLAGS: if set, the processor operates in Virtual 8086 Mode within Protected Mode.

Q250 DIFFICULT

80386 System Management Mode (SMM) is entered via:

A. RESET signal

B. Software INT

C. SMI# (System Management Interrupt) pin

D. NMI

Answer: C. SMI# (System Management Interrupt) pin
SMM is entered when SMI# pin is asserted — used for power management, hardware control, outside OS knowledge.

Q251 MODERATE FREQUENTLY ASKED

In 80386, FS and GS are:

A. Flag registers

B. Two additional segment registers added in 80386

C. General purpose registers

D. Index registers

Answer: B. Two additional segment registers added in 80386
80386 added two more segment registers FS and GS (beyond 8086's CS/DS/SS/ES) for extra data segments.

Q252 DIFFICULT FREQUENTLY ASKED

The 80386's TLB (Translation Lookaside Buffer) is used for:

A. Translating segment addresses

B. Caching recent virtual-to-physical page address translations

C. Translation of ASCII to binary

D. Storing branch history

Answer: B. Caching recent virtual-to-physical page address translations
TLB caches recent page-table translations so repeated access to same page doesn't require full page-table walk.

Pentium Processor

37 questions

Q253 EASY FREQUENTLY ASKED

The Intel Pentium processor was introduced in which year?

A 1989
B 1991
C 1993
D 1995

Answer: C. 1993
Intel Pentium (P5, 5th generation) was introduced in March 1993.

Q254 EASY FREQUENTLY ASKED

The Pentium processor features which architecture?

A Scalar
B Superscalar
C VLIW
D EPIC

Answer: B. Superscalar
Pentium uses superscalar architecture — two parallel integer execution pipes (U and V) per clock cycle.

Q255 EASY FREQUENTLY ASKED

The external data bus width of the Pentium is:

A 16-bit
B 32-bit
C 64-bit
D 128-bit

Answer: C. 64-bit
Pentium has a 64-bit external data bus — double the 80486's 32-bit bus for faster memory bandwidth.

Q256 EASY FREQUENTLY ASKED

The address bus width of the Pentium is:

A 24-bit

B 32-bit

C 36-bit

D 64-bit

Answer: B. 32-bit
Pentium has a 32-bit address bus — same as 80386/486, addressing up to 4 GB of physical memory.

Q257 MODERATE FREQUENTLY ASKED

The Pentium processor has how many integer execution pipelines?

A 1

B 2

C 4

D 8

Answer: B. 2
Pentium has 2 integer pipelines: U-pipe (any instruction) and V-pipe (simple instructions only).

Q258 MODERATE FREQUENTLY ASKED

The L1 cache in the original Pentium is:

A 4 KB unified

B 8 KB code + 8 KB data

C 32 KB unified

D 64 KB unified

Answer: B. 8 KB code + 8 KB data
Original Pentium: split L1 cache of 8 KB instruction cache + 8 KB data cache = 16 KB total.

Q259 EASY FREQUENTLY ASKED

The Pentium processor was the successor to:

A 80286
B 80386
C 80486
D 8088

Answer: C. 80486
Pentium (5th generation, P5) succeeded the Intel 80486 (4th generation) in the x86 lineup.

Q260 MODERATE FREQUENTLY ASKED

The internal register width of Pentium is:

A 16-bit
B 32-bit
C 64-bit
D 128-bit

Answer: B. 32-bit
Pentium's internal registers are 32-bit (EAX, EBX, etc.) — same as 80386/486. External data bus is 64-bit.

Q261 MODERATE FREQUENTLY ASKED

The initial clock speeds of the original Pentium were:

A 25 and 33 MHz
B 50 and 66 MHz
C 60 and 66 MHz
D 100 and 133 MHz

Answer: C. 60 and 66 MHz
The original Pentium was released at 60 MHz and 66 MHz clock speeds in 1993.

Q262 DIFFICULT

The Pentium processor contains approximately how many transistors?

A 275,000

B 1.2 million

C 3.1 million

D 5.5 million

Answer: C. 3.1 million
Original Pentium (P5) contains approximately 3.1 million transistors.

Q263 MODERATE

Pentium supports which multiprocessing configuration?

A Asymmetric only

B Symmetric Multiprocessing (SMP) — dual processor

C NUMA only

D Cluster computing

Answer: B. Symmetric Multiprocessing (SMP) — dual processors
Pentium supports SMP (Symmetric Multiprocessing), allowing dual-Pentium system configurations.

Q264 MODERATE FREQUENTLY ASKED

The Pentium integer pipeline has how many stages?

A 2

B 4

C 5

D 8

Answer: C. 5
Pentium integer pipeline: 5 stages — PF (Prefetch), D1 (Decode1), D2 (Decode2), EX (Execute), WB (Write Back).

Q265 MODERATE FREQUENTLY ASKED

MMX technology was introduced with which Pentium variant?

A. Original Pentium
B. Pentium Pro
C. Pentium MMX
D. Pentium II

Answer: C. Pentium MMX
MMX (MultiMedia eXtensions) — 57 new SIMD instructions — was introduced in Pentium MMX (1997).

Q266 MODERATE FREQUENTLY ASKED

The famous Pentium FDIV bug was related to:

A. Integer addition errors
B. Incorrect floating-point division results
C. Memory addressing fault
D. Cache coherency error

Answer: B. Incorrect floating-point division results
The Pentium FDIV bug (1994): incorrect results for certain floating-point division operations in the FPU.

Q267 MODERATE FREQUENTLY ASKED

Compared to 80386, the Pentium's external data bus is:

A. Same (32-bit)
B. Smaller (16-bit)
C. Twice as wide (64-bit)
D. Four times wider (128-bit)

Answer: C. Twice as wide (64-bit)
80386 has 32-bit external data bus; Pentium has 64-bit — doubled for higher memory bandwidth.

Q268 DIFFICULT

The Pentium Pro (P6) introduced which on-chip cache?

A. 128 KB L1

B. 256 KB or 512 KB L2 in same package

C. 32 KB L1 only

D. No cache

Answer: B. 256 KB or 512 KB L2 in same package
Pentium Pro included 256 KB or 512 KB L2 cache mounted in the same MCM package as the CPU.

Q269 DIFFICULT

The Pentium's V-pipe can only execute which type of instructions?

A. All instructions

B. Simple (short-latency) instructions only

C. Floating-point only

D. Memory operations only

Answer: B. Simple (short-latency) instructions only
The V-pipe handles only 'simple' instructions (those without dependencies/conflicts). U-pipe handles all.

Q270 DIFFICULT

Pentium's branch prediction is implemented using:

A. Branch History Table (BHT)

B. Branch Target Buffer (BTB)

C. Both BTB and BHT

D. Static prediction only

Answer: C. Both BTB and BHT
Pentium uses Branch Target Buffer (BTB) for dynamic branch prediction to reduce pipeline stalls.

Q271 DIFFICULT

The Pentium bus clock speed is what fraction of core clock?

A Equal (1:1)

B Half of core clock (1:2)

C Quarter (1:4)

D Double (2:1)

Answer: B. Half of core clock (1:2)
Pentium uses a 2:1 clock multiplier internally — core runs at 2× the external bus clock frequency.

Q272 DIFFICULT

The Pentium FPU improved performance compared to 80486 by approximately:

A 2×

B 5×

C 10×

D No improvement

Answer: B. 5×
Pentium's pipelined FPU provided approximately 5× better floating-point throughput compared to 80486.

Q273 MODERATE

Pentium Overdrive processors were used to upgrade which older machines?

A 8086-based PCs

B 80286-based PCs

C 80486-based PCs

D 80386-based PCs

Answer: C. 80486-based PCs
Pentium Overdrive chips were drop-in upgrades for 80486-based systems using the same Socket 3/2 socket.

Q274 DIFFICULT

The Pentium processor package type used in most systems was:

A. DIP-40
B. PGA-132
C. Socket 7 PGA-296
D. QFP-240

Answer: C. Socket 7 PGA-296
Most Pentium processors used Socket 5 or Socket 7 with a 296-pin SPGA (Staggered PGA) package.

Q275 MODERATE

The Pentium II used which packaging technology?

A. Socket 7
B. Slot 1 (SEC cartridge)
C. Socket 370
D. LGA 775

Answer: B. Slot 1 (SEC cartridge)
Pentium II used Slot 1 with SEC (Single Edge Contact) cartridge packaging, abandoning the socket design.

Q276 MODERATE

SSE (Streaming SIMD Extensions) was introduced in which processor?

A. Pentium MMX
B. Pentium II
C. Pentium III
D. Pentium 4

Answer: C. Pentium III
SSE with 70 new SIMD instructions was introduced in Intel Pentium III (1999).

Q277 MODERATE

The Pentium 4 used which microarchitecture with deep pipeline?

A P5

B P6

C NetBurst

D Core

Answer: C. NetBurst
Pentium 4 used Intel's NetBurst microarchitecture with a 20-stage (later 31-stage) pipeline.

Q278 DIFFICULT

The original Pentium's die size was manufactured at which process node?

A 1 micron

B 0.8 micron (800 nm)

C 0.5 micron

D 0.35 micron

Answer: B. 0.8 micron (800 nm)
The original Pentium (60/66 MHz) was manufactured at 0.8 micron (800 nm) BICMOS process.

Q279 MODERATE FREQUENTLY ASKED

Which of the following about Pentium is FALSE?

A Pentium is superscalar

B Pentium has 64-bit external data bus

C Pentium has 32-bit address bus

D Pentium internal registers are 64-bit

Answer: D. Pentium internal registers are 64-bit
Pentium's internal registers are 32-bit (same as 80386/486). Only the EXTERNAL data bus is 64-bit.

Q280 MODERATE FREQUENTLY ASKED

The Pentium can execute how many instructions per clock cycle ideally?

A. 1
B. 2
C. 4
D. 8

Answer: B. 2
Pentium's superscalar U+V pipeline can issue 2 instructions per clock cycle under ideal (pairing) conditions.

Q281 DIFFICULT

The Pentium Pro introduced which out-of-order execution concept?

A. In-order pipeline
B. Dynamic Execution (out-of-order with speculative execution)
C. VLIW
D. Barrel processing

Answer: B. Dynamic Execution (out-of-order with speculative execution)
Pentium Pro's P6 microarchitecture introduced Dynamic Execution: out-of-order execution + speculative + branch prediction.

Q282 MODERATE FREQUENTLY ASKED

The Pentium's U-pipe can execute:

A. Only simple instructions
B. Any x86 instruction (pipelined)
C. Floating-point only
D. String operations only

Answer: B. Any x86 instruction (pipelined)
U-pipe (main pipeline) can execute any x86 instruction. V-pipe handles only 'pairable' simple instructions.

Q283 DIFFICULT

Pentium Overdrive processors operated at what clock multiplier?

A. 1×

B. 2× (double speed internally)

C. 3×

D. 4×

Answer: B. 2× (double speed internally)
Pentium Overdrive chips typically ran at 2× the external bus frequency internally (clock doubling).

Q284 MODERATE FREQUENTLY ASKED

Which new instruction set extension did Pentium NOT introduce?

A. MMX

B. SSE

C. x87 FPU support

D. LOCK prefix support

Answer: B. SSE
SSE was introduced with Pentium III (1999), not original Pentium. MMX came with Pentium MMX (1997).

Q285 MODERATE FREQUENTLY ASKED

Pentium's on-chip FPU supports which floating-point standard?

A. IEEE 754

B. IBM hexadecimal float

C. BCD float only

D. Custom Intel format

Answer: A. IEEE 754
Pentium FPU is IEEE 754 compatible — supports single (32-bit), double (64-bit), and extended (80-bit) precision.

Q286 DIFFICULT

The Pentium uses which technology process?

A. 0.5 micron CMOS (Pentium at 75+ MHz)

B. NMOS 3-micron

C. Bipolar ECL

D. GaAs

Answer: A. 0.5 micron CMOS (Pentium at 75+ MHz)
Later Pentiums (75 MHz+) used 0.5 micron BiCMOS/CMOS. Original 60/66 MHz used 0.8 micron process.

Q287 DIFFICULT

The Pentium's data bus provides what peak theoretical bandwidth at 66 MHz?

A. 66 MB/s

B. 264 MB/s

C. 528 MB/s

D. 1 GB/s

Answer: C. 528 MB/s
Bandwidth = bus freq × bus width = 66 MHz × 8 bytes (64-bit) = 528 MB/s theoretical peak.

Q288 MODERATE FREQUENTLY ASKED

In Pentium's dual-pipeline execution, 'pairing' refers to:

A. Two CPUs working together

B. Two instructions executing simultaneously in U and V pipes

C. Pairing instruction and data cache

D. Branch and loop pairing

Answer: B. Two instructions executing simultaneously in U and V pipes
Instruction pairing = two compatible instructions executed in parallel in U-pipe and V-pipe per clock cycle.

Q289 DIFFICULT FREQUENTLY ASKED

The Pentium Pro introduced which advance NOT in original Pentium?

A. 64-bit data bus

B. Superscalar execution

C. Out-of-order Dynamic Execution

D. On-chip FPU

Answer: C. Out-of-order Dynamic Execution

Pentium Pro (P6, 1995) introduced Dynamic Execution (out-of-order + speculative), absent in original Pentium (P5).

Comparative Questions
39 questions

Q290 EASY FREQUENTLY ASKED

The main difference between 8086 and 8088 external data bus is:

A 8086 is 8-bit, 8088 is 16-bit

B 8086 is 16-bit, 8088 is 8-bit

C Both are 16-bit

D Both are 8-bit

Answer: B. 8086 is 16-bit, 8088 is 8-bit
8086 = 16-bit external data bus. 8088 = 8-bit external data bus. Internal architecture is identical.

Q291 MODERATE FREQUENTLY ASKED

Instruction queue size: 8086 has ___ bytes and 8088 has ___ bytes.

A 4, 6

B 6, 4

C 4, 4

D 6, 6

Answer: B. 6, 4
8086 = 6-byte prefetch queue (wider bus fetches more); 8088 = 4-byte queue (narrower 8-bit bus).

Q292 MODERATE FREQUENTLY ASKED

Compared to 8086, the 80286 has a wider address bus by how many bits?

A 2 bits

B 4 bits

C 8 bits

D 16 bits

Answer: B. 4 bits
8086: 20-bit address bus; 80286: 24-bit address bus. Difference = 4 bits more in 80286.

Q293 EASY FREQUENTLY ASKED

Which processor first introduced 32-bit registers in Intel's x86 line?

A 8086
B 80286
C 80386
D Pentium

Answer: C. 80386
80386 was Intel's first 32-bit processor, introducing 32-bit extended registers EAX, EBX, etc.

Q294 EASY FREQUENTLY ASKED

Protected Mode was first introduced in Intel x86 with:

A 8086
B 80286
C 80386
D Pentium

Answer: B. 80286
Protected Mode was first introduced in the Intel 80286 (1982).

Q295 MODERATE FREQUENTLY ASKED

Paging (hardware virtual memory) was first introduced in x86 with:

A 8086
B 80286
C 80386
D Pentium

Answer: C. 80386
Hardware paging for virtual memory was introduced for the first time in the Intel 80386.

Q296 MODERATE FREQUENTLY ASKED

Superscalar architecture in Intel x86 was first introduced with:

A 80486

B Pentium

C Pentium Pro

D 80386

Answer: B. Pentium
Pentium was Intel's first superscalar x86 processor, capable of executing 2 instructions per clock.

Q297 MODERATE FREQUENTLY ASKED

Physical memory capacities in order 8086→80286→80386→Pentium:

A 1MB→4MB→4GB→4GB

B 1MB→16MB→4GB→4GB

C 4MB→16MB→1GB→4GB

D 1MB→64MB→4GB→64GB

Answer: B. 1MB→16MB→4GB→4GB
8086=1MB(20-bit), 80286=16MB(24-bit), 80386=4GB(32-bit), Pentium=4GB(32-bit).

Q298 MODERATE FREQUENTLY ASKED

Which correctly compares 80286 and 80386?

A 80286 supports paging; 80386 does not

B 80386 is 32-bit; 80286 is 16-bit

C Both have same physical memory capacity

D 80286 has Virtual 8086 Mode

Answer: B. 80386 is 32-bit; 80286 is 16-bit
80386 is 32-bit; 80286 is 16-bit. Additionally 80386 has paging and V86 mode — 80286 has neither.

Q299 DIFFICULT FREQUENTLY ASKED

Transistor counts in order: 8086→80286→80386→Pentium:

A. 29K→134K→275K→3.1M

B. 29K→100K→200K→1M

C. 50K→134K→500K→5M

D. 10K→50K→134K→275K

Answer: A. 29K→134K→275K→3.1M
8086≈29K, 80286≈134K, 80386≈275K, Pentium≈3.1M transistors.

Q300 MODERATE FREQUENTLY ASKED

Which processor first had an on-chip FPU in x86?

A. 80386

B. 80486

C. Pentium

D. 80287

Answer: B. 80486
80486 was the first x86 processor with an integrated on-chip FPU (no separate 80487 needed).

Q301 MODERATE FREQUENTLY ASKED

Which correctly states data bus widths for these processors?

A. 8088=8, 8086=8, 80286=16, 80386=32

B. 8088=8, 8086=16, 80286=16, 80386=32

C. 8088=16, 8086=16, 80286=32, 80386=32

D. 8088=8, 8086=16, 80286=32, 80386=64

Answer: B. 8088=8, 8086=16, 80286=16, 80386=32
8088 external=8-bit, 8086=16-bit, 80286=16-bit, 80386=32-bit data bus.

Q302 MODERATE FREQUENTLY ASKED

Virtual 8086 (V86) mode was introduced in:

A. 8086
B. 80286
C. 80386
D. Pentium

Answer: C. 80386
Virtual 8086 Mode was introduced in 80386 to run multiple 8086 programs under a Protected Mode OS.

Q303 MODERATE FREQUENTLY ASKED

Assertion: 8086 can address 1 MB. Reason: 8086 has a 20-bit address bus.

A. Both true; Reason correctly explains Assertion
B. Both true; Reason does NOT explain Assertion
C. Assertion true; Reason false
D. Assertion false; Reason true

Answer: A. Both true; Reason correctly explains Assertion
Both are correct AND causally linked: 20-bit bus → 2^{20} = 1 MB. R directly explains A.

Q304 MODERATE FREQUENTLY ASKED

Which is CORRECTLY matched for year of introduction?
A. 8086 = 1978 B. 80286 = 1982 C. 80386 = 1985 D. Pentium = 1993

A. A, B, C only
B. A, B, D only
C. B, C, D only
D. A, B, C, D — all correct

Answer: D. A, B, C, D — all correct
All four are correct: 8086(1978), 80286(1982), 80386(1985), Pentium(1993).

Q305 MODERATE FREQUENTLY ASKED

Which of the following is FALSE about 8086?

A. It has 20-bit address bus

B. It has 16-bit data bus

C. It has a 4-byte instruction queue

D. It uses CISC instruction set

Answer: C. It has a 4-byte instruction queue
8086 has a 6-byte instruction queue. 8088 has 4-byte queue. The other three options are all TRUE.

Q306 DIFFICULT FREQUENTLY ASKED

Statement: Physical address FFFF0H is reached after 8086 RESET. This corresponds to CS:IP =

A. FFFF:0000H

B. 0000:FFF0H

C. FFFF:FFFFH

D. F000:FFF0H

Answer: A. FFFF:0000H
CS=FFFFH, IP=0000H → PA = FFFF0H + 0000H = FFFF0H. After RESET, CS=FFFFH, IP=0000H.

Q307 MODERATE

The 80486 (not directly in syllabus) was significant because it:

A. First 32-bit CPU

B. First with on-chip FPU and 8KB L1 cache in x86

C. First with Protected Mode

D. First superscalar

Answer: B. First with on-chip FPU and 8KB L1 cache in x86
80486 integrated FPU on-chip (previous x86s needed external co-processor) and added 8 KB L1 cache + 5-stage pipeline.

Q308 MODERATE FREQUENTLY ASKED

In terms of external data bus width: Pentium vs 80386 ratio is:

A. 1:1

B. 2:1

C. 4:1

D. 8:1

Answer: B. 2:1
80386=32-bit external data bus; Pentium=64-bit. Ratio = 64/32 = 2:1 (Pentium is twice as wide).

Q309 MODERATE FREQUENTLY ASKED

Which processor first broke the 16-bit data bus barrier in Intel x86?

A. 80286

B. 80386

C. 80486

D. Pentium

Answer: B. 80386
80386 was first with a 32-bit external data bus, breaking the 16-bit barrier of 8086/80286.

Q310 MODERATE FREQUENTLY ASKED

Which pair of processors have IDENTICAL physical memory capacity?

A. 8086 and 8088

B. 80286 and 80386

C. 80386 and Pentium

D. 8086 and 80286

Answer: C. 80386 and Pentium
80386 and Pentium both have 32-bit address buses → both can address 4 GB physical memory.

Q311 DIFFICULT

Which processor introduced hardware task switching in x86?

A 8086

B 80286

C 80386

D Pentium

Answer: B. 80286
80286 introduced hardware-assisted task switching in Protected Mode using TSS (Task State Segment).

Q312 MODERATE FREQUENTLY ASKED

The term 'segment descriptor' was introduced in which x86 processor?

A 8086

B 80286

C 80386

D Pentium

Answer: B. 80286
Segment descriptors (8-byte structures in GDT/LDT) were introduced with 80286 Protected Mode.

Q313 MODERATE FREQUENTLY ASKED

Which correctly lists processors from oldest to newest?

A 8088→8086→80286→80386→Pentium

B 8086→8088→80286→80386→Pentium

C 8086→8088→80386→80286→Pentium

D 8088→80286→8086→80386→Pentium

Answer: B. 8086→8088→80286→80386→Pentium
8086(1978)→8088(1979)→80286(1982)→80386(1985)→Pentium(1993). Chronological order.

Q314 MODERATE FREQUENTLY ASKED

The concept of 'rings' (privilege levels) was introduced in x86 with:

A. 8086
B. 80286
C. 80386
D. Pentium

Answer: B. 80286
80286 introduced privilege rings (0–3) in Protected Mode for OS security and memory protection.

Q315 DIFFICULT

Which Intel processor was NOT backward compatible with 8086 software?

A. 80286 in Real Mode
B. 8088
C. 80386 in Real Mode
D. None — all were backward compatible

Answer: D. None — all were backward compatible
All these processors (80286, 8088, 80386) maintained backward compatibility with 8086 in Real Mode.

Q316 MODERATE FREQUENTLY ASKED

Assertion: Pentium has a wider external data bus than 80386.
Reason: Pentium's 64-bit bus provides higher memory bandwidth.

A. Both true; Reason correctly explains Assertion
B. Both true; Reason does not explain
C. Assertion true; Reason false
D. Assertion false; Reason true

Answer: A. Both true; Reason correctly explains Assertion
Both correct and causally linked: 64-bit bus (vs 32-bit) directly provides higher memory bandwidth.

Q317 DIFFICULT FREQUENTLY ASKED

The total on-chip memory (L1 cache) comparison: 80486 vs Original Pentium

A 80486=8KB, Pentium=16KB

B 80486=16KB, Pentium=8KB

C Both=8KB

D Both=16KB

Answer: A. 80486=8KB, Pentium=16KB
80486 has 8 KB unified L1; Pentium has 16 KB split (8KB I + 8KB D). Pentium has 2× more L1.

Q318 MODERATE FREQUENTLY ASKED

Match processor to its address bus width:
8086→20-bit, 8088→20-bit, 80286→24-bit, 80386→32-bit, Pentium→32-bit
Which is CORRECT?

A All correct as stated

B 8086=16-bit, rest same

C 80286=32-bit, 80386=24-bit

D Pentium=64-bit

Answer: A. All correct as stated
8086=20-bit, 8088=20-bit, 80286=24-bit, 80386=32-bit, Pentium=32-bit. All as stated are correct.

Q319 MODERATE FREQUENTLY ASKED

Which processor introduced the concept of instruction queue / prefetch buffer in x86?

A 8088

B 8086

C 80286

D 80386

Answer: B. 8086
8086 introduced the 6-byte prefetch instruction queue — the first x86 processor with this feature.

Q320 MODERATE FREQUENTLY ASKED

The correct order of data bus width increase: 8088→8086→80286→80386→Pentium

A 8→8→8→16→32

B 8→16→16→32→64

C 8→16→32→32→64

D 16→16→16→32→64

Answer: B. 8→16→16→32→64
8088=8bit external, 8086=16bit, 80286=16bit, 80386=32bit, Pentium=64bit external data bus.

Q321 MODERATE FREQUENTLY ASKED

Which was NOT a feature of 8086 but WAS added in 80286?

A 16-bit registers

B Instruction queue

C Protected Mode with memory segmentation

D CISC instruction set

Answer: C. Protected Mode with memory segmentation
Protected Mode was added in 80286. 8086 has only Real Mode. The other features existed in 8086.

Q322 EASY FREQUENTLY ASKED

The term 'i386' commonly refers to:

A Intel 8086

B Intel 80386 and its 32-bit x86 architecture

C Intel 386SX only

D IBM 386

Answer: B. Intel 80386 and its 32-bit x86 architecture
i386 = Intel 80386. The term also describes the 32-bit x86 ISA (instruction set architecture) in general.

Q323 DIFFICULT

Which processor's Protected Mode is NOT backward compatible with the previous processor's Protected Mode?

A. 80386 vs 80286

B. 80286 vs 8086 (8086 has no PM)

C. Pentium vs 80386

D. All are compatible

Answer: D. All are compatible
80386 PM is backward compatible with 80286 PM (extended it). Pentium PM is compatible with 80386 PM. All are compatible

Q324 EASY FREQUENTLY ASKED

Fill in the blank: 8086 has __ address lines; this was increased to __ in 80286.

A. 16, 20

B. 20, 24

C. 24, 32

D. 20, 32

Answer: B. 20, 24
8086 has 20 address lines (1 MB); 80286 increased this to 24 address lines (16 MB).

Q325 EASY FREQUENTLY ASKED

Which of the following processors introduced the 64-bit external data bus?

A. 80386

B. 80486

C. Pentium

D. Pentium Pro

Answer: C. Pentium
Pentium was the first Intel x86 processor with a 64-bit external data bus (1993).

Q326 DIFFICULT

Which of the following is the correct full form of HMOS used in 8086 fabrication?

A. High-density Metal Oxide Semiconductor

B. High-speed NMOS

C. High-density N-channel MOS

D. High-performance MOS

Answer: C. High-density N-channel MOS
HMOS = High-density N-channel MOS — Intel's improved NMOS technology used for 8086/8088 fabrication.

Q327 MODERATE FREQUENTLY ASKED

The 80386 improved upon 80286 by adding all EXCEPT:

A. 32-bit registers

B. Paging unit

C. Virtual 8086 Mode

D. On-chip FPU

Answer: D. On-chip FPU
80386 added 32-bit architecture, paging, and V86 mode. On-chip FPU was added with 80486, not 80386.

Q328 MODERATE FREQUENTLY ASKED

Which statement about the x86 family evolution is TRUE?

A. Each generation uses completely different instruction set

B. x86 ISA maintained backward compatibility through all generations

C. 80386 cannot run 8086 code

D. Pentium uses different memory model than 80386

Answer: B. x86 ISA maintained backward compatibility through all generations
A core strength of x86: each generation maintained backward compatibility — 8086 code runs on Pentium in Real Mode.

Test

All 8 topics · Mixed difficulty · Answer key on next pages

Instructions: This is a 100-question mixed practice test covering all 8 topics. Circle your answer before checking the Answer Key at the end. Each question carries equal marks.

Q1 MODERATE

The MN/MX pin on 8088 performs the same function as on 8086, which is:

A. Memory/IO select
B. Minimum/Maximum mode select
C. Multiplexing control
D. Mode and Index select

Q2 MODERATE

The DT/R signal in 8086 minimum mode indicates:

A. Data transfer rate
B. Data Transmit (1) or Data Receive (0)
C. DMA transfer request
D. Data test result

Q3 EASY

A bus in computer architecture is:

A. A single wire
B. A group of parallel wires for signal transfer
C. A control signal only
D. A memory component

Q4 DIFFICULT

Which processor introduced hardware task switching in x86?

A. 8086
B. 80286
C. 80386
D. Pentium

Q5 MODERATE

Which correctly lists processors from oldest to newest?

A. 8088→8086→80286→80386→Pentium

B. 8086→8088→80286→80386→Pentium

C. 8086→8088→80386→80286→Pentium

D. 8088→80286→8086→80386→Pentium

Q6 DIFFICULT

Pentium's branch prediction is implemented using:

A. Branch History Table (BHT)

B. Branch Target Buffer (BTB)

C. Both BTB and BHT

D. Static prediction only

Q7 DIFFICULT

What is the purpose of the 8086 AAM instruction?

A. ASCII Adjust after Multiplication — converts binary result to BCD

B. Arithmetic Add Memory

C. Auto Address Modify

D. Aligned Address Mode

Q8 DIFFICULT

A segment descriptor in 80286 is how many bytes wide?

A. 2 bytes

B. 4 bytes

C. 8 bytes

D. 16 bytes

Q9 MODERATE

The PUSH instruction in 8086 decrements SP by:

A. 1

B. 2

C. 4

D. 8

Q10 DIFFICULT

The 80286 Local Descriptor Table (LDT) is for:

A. Global OS-level segment descriptions
B. Per-task private segment descriptions
C. Interrupt handling
D. DMA control

Q11 MODERATE

Which register pair holds the physical starting address of the stack in 8086?

A. CS:IP
B. DS:SI
C. SS:SP
D. ES:DI

Q12 MODERATE

Pentium's on-chip FPU supports which floating-point standard?

A. IEEE 754
B. IBM hexadecimal float
C. BCD float only
D. Custom Intel format

Q13 DIFFICULT

During which clock state does 8086 put the address on the bus?

A. T2
B. T3
C. T1
D. T4

Q14 MODERATE

Physical memory capacities in order 8086→80286→80386→Pentium:

A. 1MB→4MB→4GB→4GB
B. 1MB→16MB→4GB→4GB
C. 4MB→16MB→1GB→4GB
D. 1MB→64MB→4GB→64GB

Q15 MODERATE

The DEN signal in 8086 minimum mode stands for:

A. Data Enable — enables data bus transceivers

B. Decoder Enable

C. Direct Entry Node

D. Data End signal

Q16 MODERATE

80286 Real Mode segment limit is always:

A. Variable (programmable)

B. Fixed at 64 KB

C. Fixed at 16 KB

D. Fixed at 1 MB

Q17 DIFFICULT

The 8086 BIU fetches a new byte from memory whenever the queue has at least ___ empty byte(s).

A. 1

B. 2

C. 3

D. 4

Q18 MODERATE

Which 80386 mode allows protected multitasking AND backward compatibility with 8086 apps?

A. Real Mode

B. Protected Mode without paging

C. Protected Mode with Virtual 8086 Mode

D. System Management Mode

Q19 MODERATE

In 8086, the physical address is calculated as:

A. Segment + Offset

B. Segment × 16 + Offset

C. Segment ÷ 16 + Offset

D. Segment × 256 + Offset

Q20 EASY

The Pentium processor was the successor to:

A 80286
B 80386
C 80486
D 8088

Q21 EASY

The 8088 instruction set is:

A Completely different from 8086
B Identical to 8086
C A subset of 8086
D A superset of 8086

Q22 DIFFICULT

The 8086 bus operates in which mode for idle periods?

A T-idle
B T-hold
C Ti (idle state)
D T-wait

Q23 MODERATE

The clock frequency used in the original IBM PC with 8088 was:

A 2 MHz
B 4.77 MHz
C 5 MHz
D 8 MHz

Q24 EASY

The BP register in 8086 stands for:

A Base Pointer
B Bus Pointer
C Block Pointer
D Binary Pointer

Q25 MODERATE

Which CORRECTLY describes the 8088 operating modes?

A. Real Mode only

B. Protected Mode and Real Mode

C. Minimum Mode and Maximum Mode

D. Virtual Mode and Real Mode

Q26 EASY

The 8088 general-purpose registers AX, BX, CX, DX are each:

A. 8-bit

B. 16-bit

C. 32-bit

D. 4-bit

Q27 MODERATE

The 80386 operates in how many modes?

A. 2

B. 3

C. 4

D. 5

Q28 DIFFICULT

The 80286 CALLgate provides:

A. A way to call OS routines at higher privilege

B. A hardware interrupt

C. A DMA channel

D. A memory bank switch

Q29 MODERATE

Which was NOT a feature of 8086 but WAS added in 80286?

A. 16-bit registers

B. Instruction queue

C. Protected Mode with memory segmentation

D. CISC instruction set

Q30 MODERATE

The 8086 DAA instruction is used after ADD for:

A. Double addition
B. Decimal Adjust after Addition for BCD
C. Divide and Accumulate
D. Data Address Adjustment

Q31 DIFFICULT

The number of transistors in the 8088 is approximately:

A. 6,500
B. 29,000
C. 50,000
D. 134,000

Q32 DIFFICULT

80386 System Management Mode (SMM) is entered via:

A. RESET signal
B. Software INT
C. SMI# (System Management Interrupt) pin
D. NMI

Q33 EASY

The BX register in 8086 is also called:

A. Accumulator
B. Base Register
C. Counter Register
D. Data Register

Q34 MODERATE

The 8086 operates in how many modes?

A. One
B. Two
C. Three
D. Four

Q35 MODERATE

The HOLD/HLDA signals in 8086 are used for:

A. Halting CPU
B. DMA bus request/acknowledge
C. Interrupt service
D. Memory hold

Q36 MODERATE

Which of the following about 80386 is TRUE?

A. 80386 introduced Protected Mode in x86
B. 80386 has a 24-bit address bus
C. 80386 was the first Intel 32-bit x86 CPU
D. 80386 has no paging support

Q37 MODERATE

The 8086 supports which type of I/O?

A. Memory-mapped I/O only
B. Isolated I/O (port-mapped) only
C. Both memory-mapped and isolated I/O
D. Neither

Q38 MODERATE

In 8086, the BP register is used to:

A. Point to top of stack
B. Access stack frame parameters and local variables
C. Hold I/O address
D. Count loop iterations

Q39 EASY

The data bus in 8086 is:

A. Unidirectional
B. Bidirectional
C. Multiplexed only
D. Neither directional

Q40 MODERATE

80386 Linear (virtual) address space is:

A. 1 MB
B. 16 MB
C. 4 GB
D. 1 TB

Q41 MODERATE

The Overflow Flag (OF) is set when:

A. Unsigned overflow occurs
B. Signed arithmetic overflow occurs
C. Result is zero
D. Carry occurs in lower nibble

Q42 DIFFICULT

The 80286 maximum virtual memory per task in Protected Mode is:

A. 16 MB
B. 64 MB
C. 1 GB
D. 4 GB

Q43 MODERATE

The term 'segment descriptor' was introduced in which x86 processor?

A. 8086
B. 80286
C. 80386
D. Pentium

Q44 MODERATE

In 8086, the lower 16 lines AD0–AD15 are:

A. Dedicated address lines
B. Dedicated data lines
C. Multiplexed address/data lines
D. Control lines

Q45 MODERATE

Compared to 8086, the 8088 is generally:

A. Faster in data transfers

B. Slower in data transfers due to 8-bit bus

C. Same speed exactly

D. Faster due to smaller queue

Q46 MODERATE

The 8088 was used in which famous IBM PC?

A. IBM PC/AT

B. IBM PC 5150 (original IBM PC)

C. IBM PS/2 Model 30

D. IBM System/360

Q47 MODERATE

Match processor to its address bus width:
8086→20-bit, 8088→20-bit, 80286→24-bit, 80386→32-bit, Pentium→32-bit
Which is CORRECT?

A. All correct as stated

B. 8086=16-bit, rest same

C. 80286=32-bit, 80386=24-bit

D. Pentium=64-bit

Q48 MODERATE

The IBM PC/XT (1983) also used which processor?

A. 8086

B. 8088

C. 80286

D. 80386

Q49 MODERATE

The 8088's 8-bit external bus made it compatible with which older chips?

A 8085 peripherals

B Z80 chips

C 6800 chips

D Motorola 68000

Q50 DIFFICULT

In 80386 paging, a 32-bit linear address is split as:

A 10+10+12 bits (dir+table+offset)

B 12+10+10 bits

C 16+16 bits

D 20+12 bits

Q51 EASY

The Intel 80286 was introduced in which year?

A 1978

B 1980

C 1982

D 1984

Q52 DIFFICULT

80386 contains approximately how many transistors?

A 134,000

B 275,000

C 1,200,000

D 3,100,000

Q53 MODERATE

The clock speed of the original IBM PC/AT with 80286 was:

A 5 MHz

B 6 MHz

C 8 MHz

D 10 MHz

Q54 EASY

The 8086 microprocessor is a ___-bit processor.

A 8-bit

B 16-bit

C 32-bit

D 4-bit

Q55 MODERATE

The Pentium integer pipeline has how many stages?

A 2

B 4

C 5

D 8

Q56 DIFFICULT

The 80386 page directory has how many entries?

A 256

B 512

C 1024

D 4096

Q57 MODERATE

80386 was the first Intel x86 processor to support:

A Protected Mode

B Paging (hardware demand-paged virtual memory)

C Segmentation

D Hardware interrupts

Q58 MODERATE

The ALE signal in 8086 stands for:

A Address Latch Enable

B Arithmetic Logic Enable

C Address Line Enable

D Asynchronous Latch Enable

Q59 EASY

The 80286 is a ___-bit microprocessor.

A 8-bit .
B 16-bit .
C 32-bit .
D 64-bit .

Q60 DIFFICULT

How does the 8088 handle a 16-bit PUSH instruction?

A Pushes all 16 bits in one cycle using D0-D15 .
B Pushes in two separate 8-bit bus cycles .
C Converts it to an 8-bit push .
D Cannot perform 16-bit PUSH .

Q61 MODERATE

The Trap Flag (TF) in 8086 is used for:

A Enabling interrupts .
B Single-step mode debugging .
C String operation direction .
D Arithmetic overflow detection .

Q62 MODERATE

The 80286 has how many pins?

A 40 .
B 48 .
C 68 .
D 100 .

Q63 EASY

The address bus in 8086 is:

A Bidirectional .
B Unidirectional .
C Always multiplexed .
D Tristate only .

Q64 EASY

The Pentium processor features which architecture?

A Scalar

B Superscalar

C VLIW

D EPIC

Q65 EASY

The address bus width of the Pentium is:

A 24-bit

B 32-bit

C 36-bit

D 64-bit

Q66 MODERATE

In 80286 Protected Mode, segments can be how large?

A 64 KB maximum

B 16 MB maximum (24-bit base)

C Variable 1 byte to 64 KB

D Fixed 4 KB

Q67 MODERATE

In 8088, how many clock cycles does a minimum bus cycle take?

A 2

B 3

C 4

D 5

Q68 EASY

Which unit in 8086 executes instructions?

A BIU

B ALU

C EU

D MMU

Q69 DIFFICULT

Wait states (T-wait) in 8086 are inserted between which T-states?

A. T1 and T2
B. T2 and T3
C. T3 and T4
D. T4 and T1

Q70 MODERATE

The overlap of instruction fetch and execution in 8086 is called:

A. Multitasking
B. Pipelining
C. Multiplexing
D. Time-sharing

Q71 DIFFICULT

The process name for 8086 fabrication was:

A. 3-micron NMOS
B. 3-micron HMOS
C. 2-micron CMOS
D. 5-micron NMOS

Q72 DIFFICULT

The LOCK prefix in 8086 instruction set is used to:

A. Lock the keyboard
B. Assert LOCK signal to prevent other bus masters during critical section
C. Lock the interrupt
D. Lock the DMA

Q73 MODERATE

Compared to 80386, the Pentium's external data bus is:

A. Same (32-bit)
B. Smaller (16-bit)
C. Twice as wide (64-bit)
D. Four times wider (128-bit)

Q74 MODERATE

The concept of 'rings' (privilege levels) was introduced in x86 with:

A 8086

B 80286

C 80386

D Pentium

Q75 MODERATE

The improvement of 80286 over 8086 in memory capacity is:

A Same — both 1 MB

B 80286 = 16× more (16 MB vs 1 MB)

C 80286 = 4× more (4 MB vs 1 MB)

D 80286 = 256× more

Q76 MODERATE

Match segment register to its default use:
CS - Code
DS - Data
SS - Stack
ES - Extra/String destination
Which is CORRECT?

A CS-Data, DS-Code, SS-Stack, ES-Extra

B CS-Code, DS-Data, SS-Extra, ES-Stack

C CS-Code, DS-Data, SS-Stack, ES-Extra/Dest

D CS-Stack, DS-Data, SS-Code, ES-Extra

Q77 DIFFICULT

The 80386 Control Register CR0 bit 31 (PG) controls:

A Protection

B Paging unit enable/disable

C Cache

D Floating point

Q78 DIFFICULT

The Pentium Pro (P6) introduced which on-chip cache?

A. 128 KB L1
B. 256 KB or 512 KB L2 in same package
C. 32 KB L1 only
D. No cache

Q79 EASY

What does BIU stand for in 8086 architecture?

A. Binary Interface Unit
B. Bus Interface Unit
C. Basic Instruction Unit
D. Buffer Interface Unit

Q80 DIFFICULT

Which of the following is the correct full form of HMOS used in 8086 fabrication?

A. High-density Metal Oxide Semiconductor
B. High-speed NMOS
C. High-density N-channel MOS
D. High-performance MOS

Q81 EASY

The external data bus width of the Pentium is:

A. 16-bit
B. 32-bit
C. 64-bit
D. 128-bit

Q82 DIFFICULT

The Pentium Pro introduced which advance NOT in original Pentium?

A. 64-bit data bus
B. Superscalar execution
C. Out-of-order Dynamic Execution
D. On-chip FPU

Q83 DIFFICULT

The number of transistors in 8086 is approximately:

A. 6,500
B. 29,000
C. 134,000
D. 275,000

Q84 MODERATE

Which flag in 8086 is used to enable or disable maskable interrupts?

A. Trap Flag
B. Direction Flag
C. Interrupt Flag
D. Overflow Flag

Q85 MODERATE

Compared to 8086, the 80286 has a wider address bus by how many bits?

A. 2 bits
B. 4 bits
C. 8 bits
D. 16 bits

Q86 MODERATE

Protected Mode in 80286 provides which primary feature?

A. Faster execution
B. Memory protection with privilege levels
C. Floating-point support
D. Built-in graphics

Q87 MODERATE

The Auxiliary Carry Flag (AF) in 8086 is specifically used for:

A. 16-bit carry
B. BCD (Binary Coded Decimal) arithmetic corrections
C. String operations
D. Stack operations

Q88 MODERATE

The Pentium can execute how many instructions per clock cycle ideally?

A. 1
B. 2
C. 4
D. 8

Q89 MODERATE

Which correctly compares 80286 and 80386?

A. 80286 supports paging; 80386 does not
B. 80386 is 32-bit; 80286 is 16-bit
C. Both have same physical memory capacity
D. 80286 has Virtual 8086 Mode

Q90 MODERATE

Assertion: 8088 is slower than 8086 for 16-bit data operations. Reason: 8088's 8-bit external data bus requires two bus cycles for a 16-bit transfer.

A. Both A and R true; R correctly explains A
B. Both A and R true; R does NOT explain A
C. A is true; R is false
D. A is false; R is true

Q91 MODERATE

How many bytes can 8086 transfer in one bus cycle with 16-bit data bus?

A. 1 byte
B. 2 bytes
C. 4 bytes
D. 8 bytes

Q92 DIFFICULT

Statement: Physical address FFFF0H is reached after 8086 RESET. This corresponds to CS:IP =

A FFFF:0000H

B 0000:FFF0H

C FFFF:FFFFH

D F000:FFF0H

Q93 DIFFICULT

The segment:offset addressing in 8086 provides how many bytes of overlap between adjacent segments?

A None

B 16 bytes

C 256 bytes

D All of 64KB

Q94 DIFFICULT

Transistor counts in order: 8086→80286→80386→Pentium:

A 29K→134K→275K→3.1M

B 29K→100K→200K→1M

C 50K→134K→500K→5M

D 10K→50K→134K→275K

Q95 MODERATE

Which correctly states data bus widths for these processors?

A 8088=8, 8086=8, 80286=16, 80386=32

B 8088=8, 8086=16, 80286=16, 80386=32

C 8088=16, 8086=16, 80286=32, 80386=32

D 8088=8, 8086=16, 80286=32, 80386=64

Q96 DIFFICULT

The Pentium bus clock speed is what fraction of core clock?

A Equal (1:1)

B Half of core clock (1:2)

C Quarter (1:4)

D Double (2:1)

Q97 MODERATE

When the 8086 EU needs data from memory, it requests the bus from:

A ALU

B BIU

C Stack

D IP

Q98 MODERATE

Match flag to its type:
CF, ZF, SF, OF, PF, AF → Status flags
TF, IF, DF → Control flags
Which count is correct?

A 4 status, 5 control

B 6 status, 3 control

C 5 status, 4 control

D 3 status, 6 control

Q99 MODERATE

The 8086 bus cycle consists of minimum how many T-states?

A 2

B 3

C 4

D 5

Q100 DIFFICULT

The 80386 Control Register CR0 bit 0 (PE) controls:

A Paging enable

B Protection Enable (Real vs Protected mode)

C Cache enable

D Coprocessor enable

100-Question Mixed Test

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
B	**B**	**B**	**B**	**B**	**C**	**A**	**C**	**B**	**B**
Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20
C	**A**	**C**	**B**	**A**	**B**	**B**	**C**	**B**	**C**
Q21	Q22	Q23	Q24	Q25	Q26	Q27	Q28	Q29	Q30
B	**C**	**B**	**A**	**C**	**B**	**B**	**A**	**C**	**B**
Q31	Q32	Q33	Q34	Q35	Q36	Q37	Q38	Q39	Q40
B	**C**	**B**	**B**	**B**	**C**	**C**	**B**	**B**	**C**
Q41	Q42	Q43	Q44	Q45	Q46	Q47	Q48	Q49	Q50
B	**C**	**B**	**C**	**B**	**B**	**A**	**B**	**A**	**A**
Q51	Q52	Q53	Q54	Q55	Q56	Q57	Q58	Q59	Q60
C	**B**	**C**	**B**	**C**	**C**	**B**	**A**	**B**	**B**
Q61	Q62	Q63	Q64	Q65	Q66	Q67	Q68	Q69	Q70
B	**C**	**B**	**B**	**B**	**B**	**C**	**C**	**C**	**B**
Q71	Q72	Q73	Q74	Q75	Q76	Q77	Q78	Q79	Q80
B	**B**	**C**	**B**	**B**	**C**	**B**	**B**	**B**	**C**
Q81	Q82	Q83	Q84	Q85	Q86	Q87	Q88	Q89	Q90
C	**C**	**B**	**C**	**B**	**B**	**B**	**B**	**B**	**A**
Q91	Q92	Q93	Q94	Q95	Q96	Q97	Q98	Q99	Q100
B	**A**	**B**	**A**	**B**	**B**	**B**	**B**	**C**	**B**

Detailed Answers with Explanations

Q1. Answer: B. Minimum/Maximum mode select
MN/MX pin: tied HIGH (Vcc) = Minimum Mode (single CPU), tied LOW (GND) = Maximum Mode (multiprocessor). Same as 8086.

Q2. Answer: B. Data Transmit (1) or Data Receive (0)
DT/R: when HIGH = transmit (CPU sending data); when LOW = receive (CPU receiving data). Controls 8286 direction.

Q3. Answer: B. A group of parallel wires for signal transfer
A bus is a set of parallel conductors (wires/PCB traces) that carry signals between components.

Q4. Answer: B. 80286
80286 introduced hardware-assisted task switching in Protected Mode using TSS (Task State Segment).

Q5. Answer: B. 8086→8088→80286→80386→Pentium
8086(1978)→8088(1979)→80286(1982)→80386(1985)→Pentium(1993). Chronological order.

Q6. Answer: C. Both BTB and BHT
Pentium uses Branch Target Buffer (BTB) for dynamic branch prediction to reduce pipeline stalls.

Q7. Answer: A. ASCII Adjust after Multiplication — converts binary result to BCD
AAM (ASCII Adjust after Multiply) adjusts the binary result in AX after a byte multiply to unpacked BCD.

Q8. Answer: C. 8 bytes
Each segment descriptor in the 80286 GDT/LDT is 8 bytes (64 bits) wide.

Q9. Answer: B. 2
PUSH decrements SP by 2 (word = 2 bytes) before storing. The 8086 stack works with 16-bit words.

Q10. Answer: B. Per-task private segment descriptions
LDT holds per-task segment descriptors (private to each task), while GDT holds global/system-wide descriptors.

Q11. Answer: C. SS:SP
SS (Stack Segment) holds segment base; SP (Stack Pointer) holds the offset of the stack top.

Q12. Answer: A. IEEE 754
Pentium FPU is IEEE 754 compatible — supports single (32-bit), double (64-bit), and extended (80-bit) precision.

Q13. Answer: C. T1
In T1 state, 8086 puts the address on AD0–AD19 and asserts ALE to latch it externally.

Q14. Answer: B. 1MB→16MB→4GB→4GB
8086=1MB(20-bit), 80286=16MB(24-bit), 80386=4GB(32-bit), Pentium=4GB(32-bit).

Q15. Answer: A. Data Enable — enables data bus transceivers
DEN (Data Enable) enables the external 8286 data bus transceivers for data transfer.

Q16. Answer: B. Fixed at 64 KB
In 80286 Real Mode (like 8086), all segment sizes are fixed at maximum 64 KB (FFFFH offset).

Q17. Answer: B. 2
8086 BIU initiates a bus cycle to fetch 2 bytes whenever there are 2 or more empty bytes in the 6-byte queue.

Q18. Answer: C. Protected Mode with Virtual 8086 Mode
Protected Mode + Virtual 8086 Mode: each V86 task runs an 8086 program in its own protected 1 MB space.

Q19. Answer: B. Segment × 16 + Offset
Physical Address = Segment Register × 16 (or ×10H, shift left 4 bits) + Offset (Effective Address).

Q20. Answer: C. 80486
Pentium (5th generation, P5) succeeded the Intel 80486 (4th generation) in the x86 lineup.

Q21. Answer: B. Identical to 8086
8088 has an identical instruction set to 8086 — all 8086 programs run on 8088 without any modification.

Q22. Answer: C. Ti (idle state)
When no bus cycle is needed, 8086 enters Ti (idle) states — NOP cycles on the bus.

Q23. Answer: B. 4.77 MHz
The original IBM PC (5150, 1981) ran the 8088 at 4.77 MHz — derived as 3/4 of a 14.318 MHz crystal.

Q24. Answer: A. Base Pointer
BP = Base Pointer. Typically used to point to the base of the stack frame for parameter access.

Q25. Answer: C. Minimum Mode and Maximum Mode
Like 8086, 8088 has two operating modes: Minimum Mode (single CPU) and Maximum Mode (multiprocessor). No Protected Mode.

Q26. Answer: B. 16-bit
Like 8086, 8088 has four 16-bit GP registers: AX, BX, CX, DX — each splittable into H and L 8-bit halves.

Q27. Answer: B. 3
80386 has 3 operating modes: Real Mode, Protected Mode, and Virtual 8086 (V86) Mode.

Q28. Answer: A. A way to call OS routines at higher privilege
A call gate in 80286 Protected Mode allows controlled privilege escalation — user code calling OS at ring 0.

Q29. Answer: C. Protected Mode with memory segmentation
Protected Mode was added in 80286. 8086 has only Real Mode. The other features existed in 8086.

Q30. Answer: B. Decimal Adjust after Addition for BCD
DAA (Decimal Adjust Accumulator) corrects AL after ADD/ADC to maintain valid BCD representation.

Q31. Answer: B. 29,000
Intel 8088 contains approximately 29,000 transistors — the same as 8086 since the internal architecture is identical.

Q32. Answer: C. SMI# (System Management Interrupt) pin
SMM is entered when SMI# pin is asserted — used for power management, hardware control, outside OS knowledge.

Q33. Answer: B. Base Register
BX is called the Base Register and is commonly used as a memory base address pointer.

Q34. Answer: B. Two
8086 operates in two modes: Minimum Mode (single processor) and Maximum Mode (multiprocessor).

Q35. Answer: B. DMA bus request/acknowledge
HOLD: DMA requests bus. HLDA (Hold Acknowledge): CPU acknowledges, releases bus to DMA controller.

Q36. Answer: C. 80386 was the first Intel 32-bit x86 CPU
80386 was indeed the first 32-bit Intel x86 CPU (1985). It introduced paging. Protected Mode was 80286.

Q37. Answer: C. Both memory-mapped and isolated I/O
8086 supports both: memory-mapped I/O (using MOV with memory addresses) and isolated/port I/O (using IN/OUT instructions).

Q38. Answer: B. Access stack frame parameters and local variables
BP (Base Pointer) is used to access function parameters and local variables in the stack frame.

Q39. Answer: B. Bidirectional
The data bus is bidirectional — data flows TO memory (write) and FROM memory (read).

Q40. Answer: C. 4 GB
80386 provides a 32-bit linear address space = 4 GB per task (with paging, this is virtual/linear space).

Q41. Answer: B. Signed arithmetic overflow occurs
OF is set when signed arithmetic produces a result too large/small to fit — a signed overflow.

Q42. Answer: C. 1 GB
80286 Protected Mode provides up to 1 GB of virtual address space per task using segmentation.

Q43. Answer: B. 80286
Segment descriptors (8-byte structures in GDT/LDT) were introduced with 80286 Protected Mode.

Q44. Answer: C. Multiplexed address/data lines
AD0–AD15 are multiplexed: carry ADDRESS during T1 clock cycle, then DATA during T2, T3, T4.

Q45. Answer: B. Slower in data transfers due to 8-bit bus
8088 is slower than 8086 for 16-bit transfers (needs 2 cycles vs 1) but was cheaper to implement.

Q46. Answer: B. IBM PC 5150 (original IBM PC)
The original IBM PC (Model 5150, 1981) used the Intel 8088 processor at 4.77 MHz.

Q47. Answer: A. All correct as stated
8086=20-bit, 8088=20-bit, 80286=24-bit, 80386=32-bit, Pentium=32-bit. All as stated are correct.

Q48. Answer: B. 8088
IBM PC/XT (1983) used the Intel 8088 running at 4.77 MHz, same as original IBM PC.

Q49. Answer: A. 8085 peripherals
8088's 8-bit external bus allowed reuse of existing 8085-era 8-bit peripheral chips, reducing system cost.

Q50. Answer: A. 10+10+12 bits (dir+table+offset)
80386: bits 31–22 = Page Directory index (10 bits), bits 21–12 = Page Table index (10 bits), bits 11–0 = offset (12 bits).

Q51. Answer: C. 1982
Intel 80286 was introduced in February 1982.

Q52. Answer: B. 275,000
Intel 80386 contains approximately 275,000 transistors — twice as many as 80286.

Q53. Answer: C. 8 MHz
IBM PC/AT used 80286 at 8 MHz. The chip itself ran at 8, 10, 12, and 16 MHz variants.

Q54. Answer: B. 16-bit
8086 is a 16-bit microprocessor — it processes 16 bits of data at a time.

Q55. Answer: C. 5
Pentium integer pipeline: 5 stages — PF (Prefetch), D1 (Decode1), D2 (Decode2), EX (Execute), WB (Write Back).

Q56. Answer: C. 1024
80386 two-level paging: Page Directory has 1024 (2¹■) entries, each pointing to a Page Table.

Q57. Answer: B. Paging (hardware demand-paged virtual memory)
80386 was the first x86 chip to include a hardware Paging Unit for demand-paged virtual memory.

Q58. Answer: A. Address Latch Enable
ALE (Address Latch Enable) is a high pulse used to demultiplex address from AD0–AD15 lines.

Q59. Answer: B. 16-bit
80286 is a 16-bit processor — both internal registers and external data bus are 16-bit.

Q60. Answer: B. Pushes in two separate 8-bit bus cycles
Since the bus is only 8-bit, the 8088 performs 16-bit PUSH in two consecutive 8-bit bus cycles — high byte then low byte.

Q61. Answer: B. Single-step mode debugging
TF=1 causes 8086 to enter single-step mode — an internal interrupt after each instruction for debugging.

Q62. Answer: C. 68
80286 has 68 pins in a PGA (Pin Grid Array) package, compared to 8086's 40 pins.

Q63. Answer: B. Unidirectional
The address bus is unidirectional — addresses flow only FROM the CPU to memory/I/O devices.

Q64. Answer: B. Superscalar
Pentium uses superscalar architecture — two parallel integer execution pipes (U and V) per clock cycle.

Q65. Answer: B. 32-bit
Pentium has a 32-bit address bus — same as 80386/486, addressing up to 4 GB of physical memory.

Q66. Answer: B. 16 MB maximum (24-bit base)
In Protected Mode, 80286 segments: base is 24-bit (16 MB range) with 16-bit limit up to 64 KB per segment.

Q67. Answer: C. 4
Like 8086, a minimum 8088 bus cycle takes 4 T-states (T1-T4), but 16-bit data needs two such cycles.

Q68. Answer: C. EU
EU (Execution Unit) decodes and executes instructions taken from the instruction queue.

Q69. Answer: C. T3 and T4
Wait states are inserted between T3 and T4 when READY signal is low, extending the bus cycle.

Q70. Answer: B. Pipelining
BIU fetching next instruction while EU executes current one is called pipelining — increases throughput.

Q71. Answer: B. 3-micron HMOS
Intel 8086 was originally fabricated in HMOS (High-density N-channel MOS) 3-micron process technology.

Q72. Answer: B. Assert LOCK signal to prevent other bus masters during critical section
LOCK prefix asserts the LOCK pin, preventing other bus masters (in max mode) from acquiring the bus.

Q73. Answer: C. Twice as wide (64-bit)
80386 has 32-bit external data bus; Pentium has 64-bit — doubled for higher memory bandwidth.

Q74. Answer: B. 80286
80286 introduced privilege rings (0–3) in Protected Mode for OS security and memory protection.

Q75. Answer: B. 80286 = 16× more (16 MB vs 1 MB)
8086 = 1 MB (20-bit bus); 80286 = 16 MB (24-bit bus). 16 MB / 1 MB = 16× improvement.

Q76. Answer: C. CS-Code, DS-Data, SS-Stack, ES-Extra/Dest
CS=Code segment (instructions), DS=Data segment (default data), SS=Stack segment, ES=Extra (string dest).

Q77. Answer: B. Paging unit enable/disable
CR0.PG=1 enables the paging unit. CR0.PG=0 disables paging (flat physical memory used).

Q78. Answer: B. 256 KB or 512 KB L2 in same package
Pentium Pro included 256 KB or 512 KB L2 cache mounted in the same MCM package as the CPU.

Q79. Answer: B. Bus Interface Unit
BIU = Bus Interface Unit. It manages all bus operations: fetching instructions and transferring data.

Q80. Answer: C. High-density N-channel MOS
HMOS = High-density N-channel MOS — Intel's improved NMOS technology used for 8086/8088 fabrication.

Q81. Answer: C. 64-bit
Pentium has a 64-bit external data bus — double the 80486's 32-bit bus for faster memory bandwidth.

Q82. Answer: C. Out-of-order Dynamic Execution
Pentium Pro (P6, 1995) introduced Dynamic Execution (out-of-order + speculative), absent in original Pentium (P5).

Q83. Answer: B. 29,000
Intel 8086 contains approximately 29,000 transistors.

Q84. Answer: C. Interrupt Flag
The Interrupt Flag (IF): IF=1 enables, IF=0 disables maskable hardware interrupts (INTR pin).

Q85. Answer: B. 4 bits
8086: 20-bit address bus; 80286: 24-bit address bus. Difference = 4 bits more in 80286.

Q86. Answer: B. Memory protection with privilege levels
Protected Mode provides hardware-enforced memory protection, privilege levels (rings 0–3), and virtual memory.

Q87. Answer: B. BCD (Binary Coded Decimal) arithmetic corrections
AF is set when carry/borrow occurs from bit 3 to bit 4 — used by DAA/DAS for BCD arithmetic.

Q88. Answer: B. 2
Pentium's superscalar U+V pipeline can issue 2 instructions per clock cycle under ideal (pairing) conditions.

Q89. Answer: B. 80386 is 32-bit; 80286 is 16-bit
80386 is 32-bit; 80286 is 16-bit. Additionally 80386 has paging and V86 mode — 80286 has neither.

Q90. Answer: A. Both A and R true; R correctly explains A
Both correct and causally linked: 8-bit bus needs 2 cycles per 16-bit word, making it slower than 8086's single cycle.

Q91. Answer: B. 2 bytes
8086 transfers 2 bytes (1 word = 16 bits) per bus cycle using its 16-bit data bus.

Q92. Answer: A. FFFF:0000H
CS=FFFFH, IP=0000H → PA = FFFF0H + 0000H = FFFF0H. After RESET, CS=FFFFH, IP=0000H.

Q93. Answer: B. 16 bytes
Adjacent segments (offset by 1 paragraph = 16 bytes) overlap by 64KB - 16 bytes. Adjacent = 16 bytes different start.

Q94. Answer: A. 29K→134K→275K→3.1M
8086≈29K, 80286≈134K, 80386≈275K, Pentium≈3.1M transistors.

Q95. Answer: B. 8088=8, 8086=16, 80286=16, 80386=32
8088 external=8-bit, 8086=16-bit, 80286=16-bit, 80386=32-bit data bus.

Q96. Answer: B. Half of core clock (1:2)
Pentium uses a 2:1 clock multiplier internally — core runs at 2× the external bus clock frequency.

Q97. Answer: B. BIU
EU requests the BIU to perform memory read/write operations — EU itself has no direct bus access.

Q98. Answer: B. 6 status, 3 control
6 status flags: CF, ZF, SF, OF, PF, AF. 3 control flags: TF (Trap), IF (Interrupt), DF (Direction).

Q99. Answer: C. 4
A minimum 8086 bus cycle has 4 T-states: T1 (address), T2 (data setup), T3 (data valid), T4 (complete).

Q100. Answer: B. Protection Enable (Real vs Protected mode)
CR0.PE=0: Real Mode; CR0.PE=1: Protected Mode. Setting PE switches the 80386 to Protected Mode.